



NanoPhotometer™ USER MANUAL

Version 2.0





Implen GmbH · Schatzbogen 52 · D-81829 Germany

Declaration of conformity for the NanoPhotometer™

This is to certify that the Implen NanoPhotometer™ conforms to the requirements of the following Directives:

73/23/EEC & 89/336/EEC

Standards, to which conformity is declared, where relevant, are as follows

EN 61010-1: 2001 Safety requirements for electrical equipment for measurement, control and laboratory use.

EN 61326-2.3: 1998 Electromagnetic compatibility - generic emission standard Electrical equipment for measurement, control and laboratory use.

EN 61000-4-6: 1992 Electromagnetic compatibility - generic immunity standard part 1. Residential, commercial and light industry.

For further information, including unpacking, positioning and installation of the products please refer to the user manual.

Signed:

Dated: June 1, 2006

Dr. Thomas Sahiri
Managing Director
Implen GmbH



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1. ESSENTIAL SAFETY NOTES

There are a number of warning labels and symbols on your instrument. These are there to inform you where potential danger exists or particular caution is required. Before commencing installation, please take time to familiarise yourself with these symbols and their meaning.



Caution (refer to accompanying documents).
Background colour yellow, symbol and outline black.

1.1 Unpacking, Positioning and Installation

- Check the contents of the pack against the packing list. If any shortages are discovered, inform your supplier immediately.
- Inspect the instrument for any signs of damage caused in transit. If any damage is discovered, inform your supplier immediately.
- Ensure your proposed installation site conforms to the environmental conditions for safe operation:
 - Indoor use only.
 - Temperature range 5°C to 35°C. Note that if you use the instrument in a room subjected to extremes of temperature change during the day, it may be necessary to recalibrate (by switching off and then on again) once thermal equilibrium has been established (2-3 hours).
 - Maximum relative humidity of 80% up to 31°C decreasing linearly to 50% at 40°C.
- The instrument must be placed on a stable, level bench or table that can take its weight (< 4.5 kg) so that air can circulate freely around the instrument.
- This equipment must be connected to the power supply with the power cord supplied. It can be used on 90 – 240 V, 50-60 Hz supplies.
- If the instrument has just been unpacked or has been stored in a cold environment, it should be allowed to come to thermal equilibrium for 2-3 hours in the laboratory before switching. This will prevent calibration failure as a result of internal condensation.
- Switch on the instrument via the keypad () after it has been plugged in. The instrument will perform a series of self-diagnostic checks.
- Please read through this user manual prior to use.
- Please contact your original supplier in the first instance if you experience technical or sample handling difficulties.

If this equipment is used in a manner not specified or in environmental conditions not appropriate for safe operation, the protection provided by the equipment may be impaired and instrument warranty withdrawn.

2. INTRODUCTION

2.1 Your spectrophotometer

Your spectrophotometer is a simple-to-use UV/Visible instrument with a CCD array detector (1024 pixels). It has no moving parts, which is the basis of the rapid scanning operating system.

The user interface is built around folders which are displayed on the home page when the instrument is switched on. Different folders are numbered and opened by using the associated number key on the keypad. After switch on and calibration, the default home page is "NanoPhotometer™" offering the choice of:

NanoPhotometer™	
 LabelGuard Applications	
 Cuvette Applications	
 Functions	
 User Methods	
 Utilities	
Key pad number	Description
1	Life Science methods such as nucleic acid assays and protein assays using the LabelGuard™ Microliter Cell
2	Life Science methods such as nucleic acid assays, protein assays and cell density
3	General spectroscopic methods
4	Contains nine folders that can store user adapted methods
5	Instrument set up (date, time, number format), and printer and Baseline Compensation set up.

The instrument is equipped with a standard USB port. The optional NanoPhotometer™ Software Package (NP-SW100) is necessary to connect the NanoPhotometer™ to a PC. The software enables the user to "print through" the PC directly to the printer that is connected to it. Data may be stored as Excel spreadsheet, EMF graphics file, a comma delimited (csv) data file, a tab delimited (txt) data file or in native NanoPhotometer™ Software format for later access.

Alternatively, results may be saved on a SD Memory Card or sent to the PC via a Bluetooth accessory; these can either be supplied pre-installed or are available as an optional accessory if the need for the use arises after installation of the product. The NanoPhotometer™ Software works in a similar way.

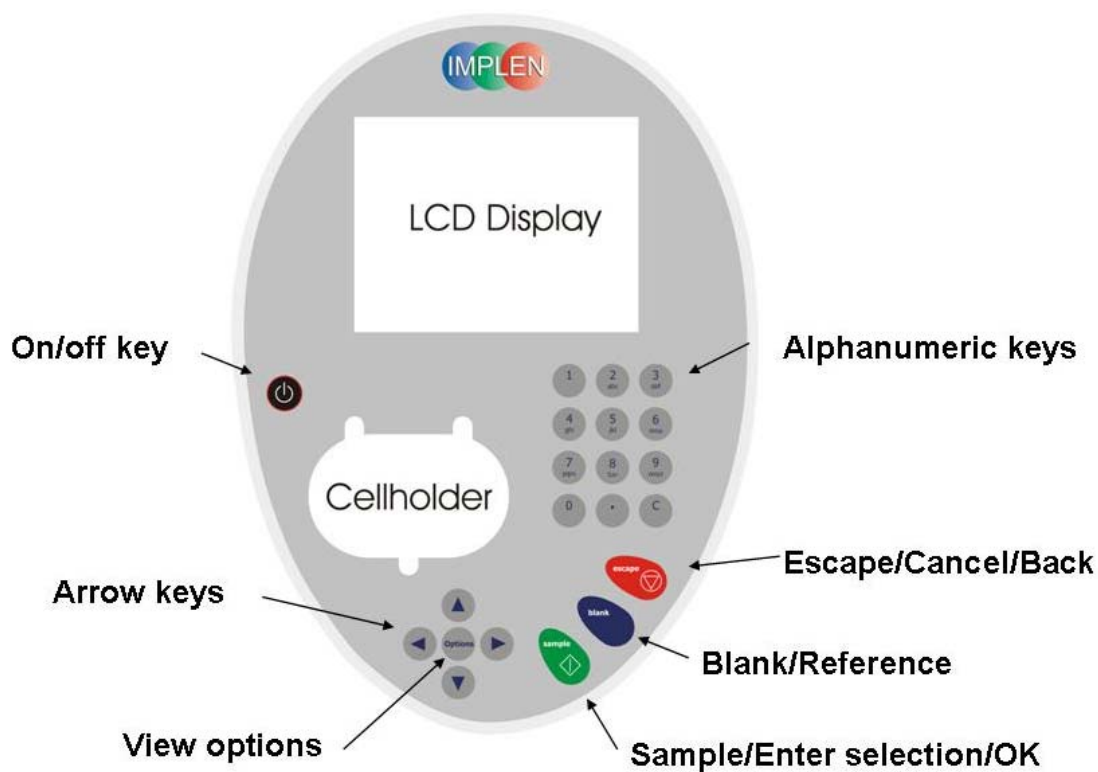
A printer is available for the instrument; this may either be supplied pre-installed or is available as an optional accessory if the need for its use arises after installation of the product.

2.2 Sample handling tips

- Note that the light beam is directed from RIGHT to LEFT through the cell chamber; therefore please ensure the measurement cell is inserted in the correct alignment.
- Insert the measurement cell always in the same direction.
- The cell holder supplied with the instrument accepts the LabelGuard™ Microliter Cell and standard 10 mm pathlength quartz, glass or plastic cells.
- The optical height of the NanoPhotometer™ is 15 mm.
- The minimum volume that can be used is 0.7 µl with the LabelGuard™ Microliter Cell.
- 12 mm test tubes may be used (e.g. for cell cultures), however they are not recommended as higher quality data is produced by using disposable cuvettes for the analysis. If used, align the indicator line on 12 mm test tubes in the same direction to ensure reproducible positioning of the tube. Note that test tubes do not last forever, and that the surface becomes scratched and blemished through repetitive use; if this is the case they should be replaced.

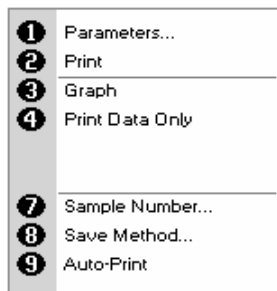
2.3 Keypad and display

The back-lit liquid crystal display is very easy to navigate around using the alphanumeric entry and navigation arrow keys on the hard wearing, spill proof membrane keypad.



Key	Action
On/off key	Turns the instrument on/off.
Arrow keys	Use the four arrow keys to navigate around the display and select the required setting from the active (highlighted) option.
View Options	View options for that application mode. Some of these are common to all applications and described below. Options unique to an application are described in the relevant section.
Alphanumeric keys	Use these to enter parameters and to write text descriptions where appropriate, or required. Use repeated key presses to cycle through lower case, number and upper case. Leave for 1 second before entering next character. Use C button to backspace and 1 to enter a space.
Escape/Cancel/Back: 	Escape from a selection and return to the previous folder. Cancel a selection. Stop making measurements.
Blank/Reference	Set reference to 0.000 A or 100%T on a reference solution at the current wavelength in the mode selected. When in scan mode, does a reference scan.
Sample/Enter selection/Back: 	Enter, or confirm a selection. Take a measurement.

Options (select using key pad numbers)



Options (select using key pad numbers)

1. View parameters for the experiments.
2. Print the results.
- 3., 4. Is described in 4. LabelGuard Applications and Cuvette Applications.
7. Define the sample number you wish to start from.
8. Save the parameters as a method to a defined folder name with a defined method name.
9. Toggle auto-print on/off. Default is off.

Exit options by pressing Escape , or wait.

Experienced operators can use the numeric keys as a shortcut to the option required without needing to enter the Options menu.

3. THE LABELGUARD™ MICROLITER CELL

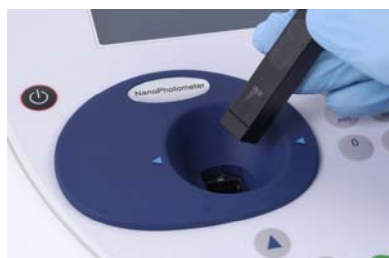
With its innovative optical pathway the cell is designed for optimum measurement results with submicroliter sample volumes ranging from 0.7 µl up to 10 µl of undiluted sample. Due to a pathlength of 0.1 mm, 0.2 mm, 1 mm and 2 mm the cell is offering an automatic dilution of 1/100, 1/50, 1/10 and 1/5 in comparison to a standard cuvette measurement. Because the measurements are processed with undiluted samples, the reproducibility of the results is extremely high. If desired, samples can be retrieved after the measurement for further processing. The LabelGuard™ Microliter Cell can be used for all UV/Vis analysis utilizing the wavelength range of 190 to 1100 nm.

The LabelGuard™ Microliter Cell is delivered with two lids for 0.2 mm (Lid 50) and 1 mm (Lid 10) pathlength, which cover most applications. Lid 5 (2 mm pathlength) and Lid 100 (0.1 mm pathlength) are optional. The dilution factor (lid factor) is printed on the lid. Please make sure that you use the appropriate lid for your sample.



Note: The LabelGuard™ Microliter Cell is delivered with two adapters. The centre height can be adjusted to 8.5 mm (no adapter), 15 mm (Z15) or 20 mm (Z20). Before starting make sure that the centre height of the LabelGuard™ is adjusted to the right setting. For the NanoPhotometer the value is 15 mm!

3.1 Technical instructions



Step 1 Insert the LabelGuard™ Microliter Cell into the cell holder with the cell windows facing the light beam. The light beam is directed from RIGHT to LEFT as indicated with small blue arrows. Insert the LabelGuard™ Microliter Cell always in the same direction.

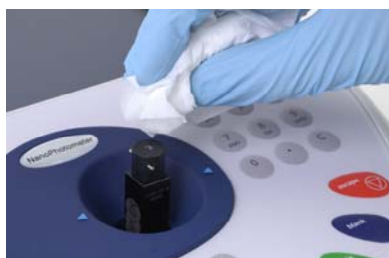


Step 2 Pipette the appropriate sample volume onto the centre of the measuring window. **Warning!!** Do not overfill the well.

Lid	Sample volume	Pathlength	Dilution
100 (optional)	0.7 – 4 µl	0.1 mm	1:100
50	0.7 – 4 µl	0.2 mm	1:50
10	3 – 5 µl	1 mm	1:10
5 (optional)	6 – 10 µl	2 mm	1:5



Step 3 Make sure that the lid fits exactly for the measurements onto the positioning supports mounted to the body of the cell. Take measurement. **Remember** to consider the lid factor in your instrument software. Please refer to 3.2 Software instructions for detailed information.



Step 4 Take lid off and retrieve sample with a pipette for further applications if desired. Remove sample residues from the measurement window and the lid mirror. Clean measurement window and lid mirror well with a fluff free tip or laboratory wipe. Use water, ethanol or isopropanol. Do not use aggressive solvents like strong acids or bases or organic solvents at any time.

Important Note: Residual fluffs must be removed for optimum performance - use dry pressurized air (oil free) if needed.

Your cell is ready for the next sample.



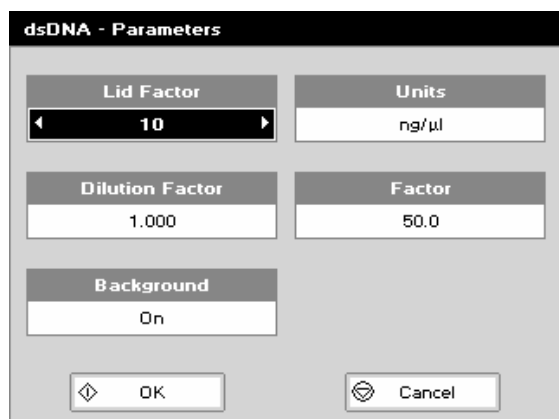
Operation Limitations: Do not autoclave the unit! Do not use an ultrasound bath to clean! Do not drop in water or solvent bath. The unit is water resistant, but not water proof!

3.2 Software instructions

The **LabelGuard Applications** and **Cuvette Applications** are very similar concerning the analysis of dsDNA, ssDNA, RNA, Oligonucleotides, protein UV and protein dye analysis. This section describes the specific features which have to be considered using the LabelGuard™ Microliter Cell. For general information please follow the detailed instructions under LabelGuard Applications and Cuvette Applications.

The procedure is as follows:

Exemplary Parameter Screen



Parameter Screen

- Step 1** Press 1 to select **LabelGuard Applications** folder
- Step 2** Press 1 to select **Nucleic Acids** folder OR 2 to select **Protein** folder.
- Step 3** Select the method you want to use by pressing the corresponding number.
- Step 4** Select the **Lid Factor** using the left and right arrows.

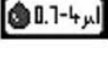
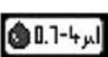
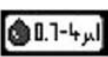
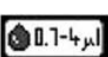
Lid	Lid Factor	Dilution
100 (optional)	100	1:100
50	50	1:50
10	10	1:10
5 (optional)	5	1:5

- Step 5** Select subsequent parameters and specifications as described under 4. **LabelGuard Applications and Cuvette Applications**.

After the selections are confirmed the results screen displays in top left corner the chosen Lid and the required sample volume.

Important Information:

If the absorbance value of the sample is not between **0.02** and **1.7** the following **“Warning message”** and **“Instruction”** will be displayed in the top left corner of the result screen.

Lid	Required volume	Warning message	Instruction
 optional		 Abs too low	Sample concentration is too low
 optional		 Abs is too high	 change to lid 
 		 Abs too low	Sample concentration is too low (or change to lid 5 if available)
 		 Abs is too high	 change to lid 
 		 Abs too low	 change to lid 
 		 Abs is too high	 Physical dilution of the sample is necessary (or change to lid 100 if available)
 optional		 Abs too low	 change to lid 
 optional		 Abs is too high	 Physical dilution of the sample is necessary

4. LABELGUARD APPLICATIONS AND CUVETTE APPLICATIONS

The NanoPhotometer offers a complete solution for submicroliter volume and standard volume applications. With the LabelGuard™ Microliter Cell the required sample volume ranges from 0.7 µl to a maximal sample volume of 10 µl. Standard volume applications can be performed with 10 mm pathlength quartz, glass or plastic cuvettes.

Note:

Within the Utilities folder the user has the possibility to select various options that define data out-put. Before starting it is advisable to check that Printer Options have been appropriately set for your experiment. Please refer to 7.3 Printer.

The **LabelGuard Applications** folder and the **Cuvette Applications** folder contain different sub folders: **Nucleic Acids**, **Protein** and **OD 600** (Cell Density). Contents of these sub folders are detailed below.

Folder	Application	Recommended Measurement Cell
Nucleic Acids		
DNA	Concentration, purity check and dye incorporation for DNA samples	LabelGuard / Cuvette
RNA	Concentration, purity check and dye incorporation for RNA samples	LabelGuard / Cuvette
Oligo	Concentration, purity check and dye incorporation for oligo samples	LabelGuard / Cuvette
Protein		
Protein UV (Christian-Warburg)	Protein determination at 280 nm	LabelGuard / Cuvette
Protein Dye	Protein determination at 280 nm and dye incorporation	LabelGuard / Cuvette
BCA	Protein determination at 562 nm	Cuvette
Bradford	Protein determination at 595 nm	Cuvette
Lowry	Protein determination at 750 nm	Cuvette
Biuret	Protein determination at 546 nm	Cuvette
Cell Count		
OD600	Cell density at 600 nm	Cuvette

4.1 Characterization of DNA, RNA and Oligonucleotides

4.1.1 General Information

Nucleic Acid Quantification (NAQ)

- Nucleic acids can be quantified at 260 nm because it is well established that a solution of dsDNA in a 10 mm pathlength cell with an optical density of 1.0 has a concentration of 50 µg/ml, ssDNA of 37 µg/ml or 40 µg/ml in the case of RNA. Oligonucleotides have a corresponding factor of 33 µg/ml, although this does vary with base composition; this can be calculated if the base sequence is known. Please refer to 11.1 Nucleic acid quantification for further details.
- The instrument uses factors 50, 37, 40 and 33 as default settings for dsDNA, ssDNA, RNA and Oligonucleotides, respectively, and compensation factors for dilution and use of cells which do not have 10 mm pathlength. Dilution factor and cell pathlength can be entered.

Nucleic Acid Purity Checks

- Nucleic acids extracted from cells are accompanied by protein, and extensive purification is required to separate the protein impurity. The 260/280 ratio gives an indication of purity; it is only an indication, however, and not a definitive assessment. Pure DNA and RNA preparations have expected ratios of ≥ 1.8 and ≥ 2.0 , respectively; deviations from this indicate the presence of impurity in the sample, but care must be taken in interpretation of results.
- The 260 nm reading is taken near the top of a broad peak in the absorbance spectrum for nucleic acids, whereas the 280 nm reading is taken on a steep slope (i.e. small changes in wavelength cause large changes in absorbance). Consequently, small variations in wavelength at 280 nm will have a greater effect on the 260/280 ratio than variations will at 260 nm. Thus different instruments of the same and different types may give slightly different ratios due to variations in wavelength accuracy. But each instrument will give consistent results within itself.
- Concentration also affects 260/280 readings. If a solution is too dilute, the readings will be at the instrument's detection limit, and results may vary as there is less distinction of the 260 peak and the 280 slope from the background absorbance. This is one reason why the Abs 260 value should be greater than 0.1 for accurate measurements.
- An elevated absorbance at 230 nm can indicate the presence of impurities as well; 230 nm is near the absorbance maximum of peptide bonds and also indicates buffer contamination since TRIS, EDTA and other buffer salts absorb

at this wavelength. When measuring RNA samples, the 260/230 ratio should be > 2.0; a ratio lower than this is generally indicative of contamination with guanidinium thiocyanate, a reagent commonly used in RNA purification and which absorbs over the 230 - 260 nm range. A wavelength scan of the nucleic acid is particularly useful for RNA samples.

- The instrument can display 260/280 and 260/230 ratios, and compensates for dilution and use of cells that do not have 10 mm pathlength; dilution factor and cell pathlength can be entered.

Fluorescent dye incorporation

- To determine the dye incorporation rate, the absorbance reading at the wavelength reported for maximum absorbance of the fluorescence dye is used. The corresponding extinction coefficient of the dye is used in the Lambert-Beer Law to determine the dye concentration ($c = A / (e * d)$). Comparing these values with the DNA concentration gives a dye incorporation rate. For further details please refer to 11.2 Nucleic acid fluorescent dye incorporation.

Use of Background Correction

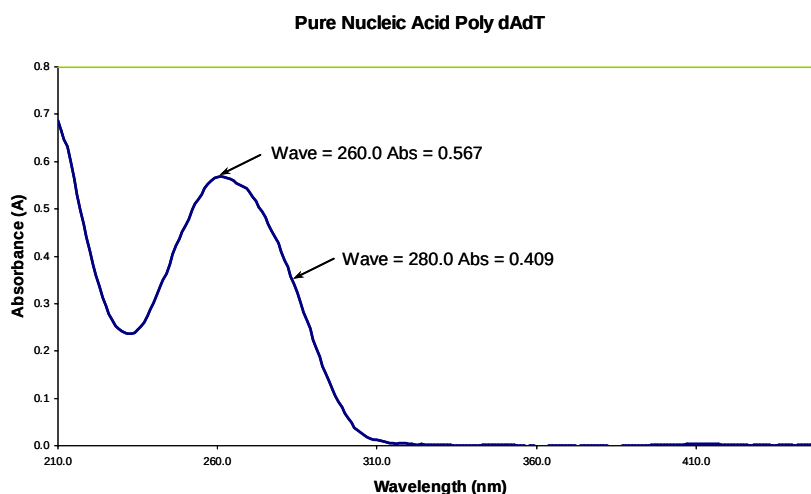
- Background correction at a wavelength totally separate from the nucleic acid and protein peaks at 260 and 280 nm, respectively, is sometimes used to compensate for the effects of background absorbance. The wavelength used is 320 nm and it can allow for the effects of turbidity, high absorbance buffer solution and the use of reduced aperture cells. The instrument can use background correction.
- If it is used, there will be different results from those when unused, because Abs320 is subtracted from Abs260 and Abs280 prior to use in equations:

$$\text{Concentration} = (\text{Abs } 260 - \text{Abs } 320) * \text{Factor}$$

$$\text{Abs ratio} = (\text{Abs } 260 - \text{Abs } 320) / (\text{Abs } 280 - \text{Abs } 320)$$

$$\text{Abs ratio} = (\text{Abs } 260 - \text{Abs } 320) / (\text{Abs } 230 - \text{Abs } 320)$$
- If your laboratory has not used background correction before, set this option to NO.
- The use of background correction can remove variability due to handling effects of low volume disposable cells.

Spectral scan of nucleic acid



Note:

- absorbance maximum near 260 nm and absorbance minimum near 230 nm
- flat peak near 260 nm and steep slope at 280 nm
- very little absorbance at 320 nm

Operation of the instrument for Nucleic Acid measurements is described in the following sections.

DNA and RNA are very similar, whilst in Oligo it is possible to calculate the factor from the composite bases by entering the proportions of the 4 bases.

4.1.2 Analysis of dsDNA, ssDNA and RNA

The procedure is as follows:

Parameter Screen

LabelGuard Applications

dsDNA - Parameters

Lid Factor	Units
10	ng/μl
Dilution Factor	Factor
1.000	50.0
Background	
On	
OK	Cancel

Cuvette Applications

dsDNA - Parameters

Pathlength	Units
10 mm	ng/μl
Dilution Factor	Factor
1.000	50.0
Background	
On	
OK	Cancel

Results Screen

dsDNA

A230	0.076 A	Sample 1
A260	0.119 A	
A280	0.046 A	
A320	0.000 A	
A260/A280	2.587	Concentration 59.5
A260/A230	1.566	Units μg/ml

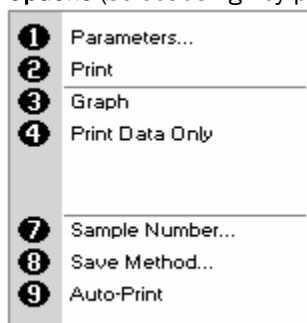
Parameter Screen

- Step 1** Press 1 for **LabelGuard** OR 2 for **Cuvette** folder
- Step 2** Press 1 to select **Nucleic Acids** folder
- Step 3** Press 1 to select **dsDNA** mode OR 2 to select **ssDNA** mode OR 3 to select **RNA** mode
- Step 4** Using the LabelGuard™ Applications select the **Lid Factor** as described under 3.2. Using Cuvette Applications select **Pathlength** using the left and right arrows. Options are 5 mm or 10 mm.
- Step 5** Enter the **Dilution Factor** using the keypad numbers. Range 1.00 to 9999. Use the C button to backspace and clear the last digit entered OR press **Options** to enter the dilution factor screen. Enter the volume of the sample using the keypad numbers. Range 0.01 to 9999. Enter the volume of the diluent using the keypad numbers. Range 0.01 to 9999. Press to calculate the dilution factor and return to the Parameters screen OR Press Cancel to cancel the selections and return to the Parameters screen.
- Step 6** **Background** correction at 320 nm is recommended to be switched on.
- Step 7** Select the **Units** of measurement using the left and right arrows. Options: μg/ml, ng/μl, μg/μl.
- Step 8** Enter the **Factor** using the keypad numbers. Default value is 50 for dsDNA, 37 for ssDNA and 40 for RNA, range is 0.01 to 9999.
- Step 9** Press OK to enter the Results screen OR Cancel to return to the **Nucleic Acids** folder.

Results Screen

- Step 10** Insert the reference sample. Press **Blank** Key. This will be used for all subsequent samples until changed.
- Step 11** Insert sample and press . This measures at the selected wavelengths and displays the results. The sample concentration, the ratio of A260/A280 and A260/A230 are calculated (corrected by the background wavelength value if selected).
- Step 12** If the absorbance value of the sample is not between 0.02 and 1.7 a **“Warning message”** and **“Instruction”** will be displayed in the top left corner of the result screen. Please refer to 3.2 Software instructions for further information.
- Step 13** Repeat for all samples.
- Step 14** Press **Options** to display available Options which are described below.
- Step 15** Press and confirm with to return to the **Nucleic Acids** folder.

Options (select using key pad numbers)



Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. The graph shows a wavescan plot across the range 220 nm to 750 nm with cursors denoting 230, 260, 280 and (if background correction selected) 320 nm.
- 4) Toggle on/off to display the data table only in the print-out.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.
Exit options by pressing **0**, **OR** wait.

4.1.3 Analysis of Oligonucleotides

The procedure is as follows:

Parameter Screen

LabelGuard Applications

Oligo - Parameters

Lid Factor	10	Units	pmol/μl
Dilution Factor	1.000	A	10
		C	10
		G	10
		T	10
Background	On		

OK Cancel

Cuvette Applications

Oligo - Parameters

Pathlength	10 mm	Units	pmol/μl
Dilution Factor	1.000	A	10
		C	10
		G	10
		T	10
Background	On		

OK Cancel

Results Screen

Oligo

A230	0.076 A	Sample	1
A260	0.119 A	Concentration	2.579
A280	0.047 A		
A320	0.009 A		
A260/A280	2.895	Units	pmol/μl
A260/A230	1.642		

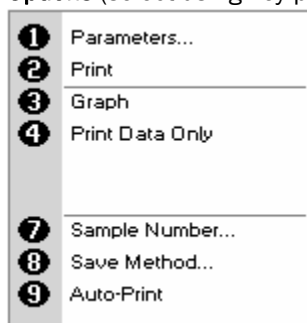
Parameter Screen

- Step 1** Press 1 for **LabelGuard** OR 2 for **Cuvette** folder.
- Step 2** Press 1 to select **Nucleic Acids** folder.
- Step 3** Press 4 to select **Oligo** mode.
- Step 4** Using the LabelGuard™ Applications select the **Lid Factor** as described under 3.2. Using Cuvette Applications select **Pathlength** using the left and right arrows. Options are 5 mm or 10 mm.
- Step 5** Enter the **Dilution Factor** using the keypad numbers. Range 1.00 to 9999. Use the C button to backspace and clear the last digit entered OR press **Options** to enter the dilution factor screen. Enter the volume of the sample using the keypad numbers. Range 0.01 to 9999. Enter the volume of the diluent using the keypad numbers. Range 0.01 to 9999. Press $\diamond$ to calculate the dilution factor and return to the Parameters screen OR Press Cancel ∇ to cancel the selections and return to the Parameters screen.
- Step 6** **Background** correction at 320 nm is recommended to be switched on.
- Step 7** Select the **Units** of measurement using the left and right arrows. Options: μg/ml, ng/μl, μg/μl and pmol/μl.
- Step 8** Enter the **Factor** using the keypad numbers. Default value is 33, range is 0.01 to 9999.
- Step 9** If pmol/μl is selected there are two options to set the factor
1. A selection table denoting the ratios of the 4 bases according to the oligo sequence. Enter the proportions of bases present using the keypad numbers and up and down arrows to move between boxes. Default is 10 for each, range is 0 to 9999.
 2. Enter the known extinction factor of the oligo used: factor range 0.01 to 9999 for ratio = $[1 / \text{extinction coefficient} * 10^{-6}]$.
- Step 10** Press OK $\diamond$ to enter the Results screen OR Cancel ∇ to return to the **Nucleic Acids** folder.

Results Screen

- Step 11** Insert the reference sample. Press **Blank** Key. This will be used for all subsequent samples until changed.
- Step 12** Insert sample and press $\diamond$. This measures at the selected wavelengths and displays the results. The sample concentration and the ratio of A260/A280 and A260/A230 are calculated (corrected by the background wavelength value if selected).
- Step 13** If the absorbance value of the sample is not between 0.02 and 1.7 a **Warning message** and **Instruction** will be displayed in the top left corner of the result screen. Please refer to 3.2 Software instructions for further information.
- Step 14** Repeat for all samples.
- Step 15** Press **Options** to display available Options which are described below.
- Step 16** Press ∇ and confirm with $\diamond$ to return to the **Nucleic Acids** folder.

Options (select using key pad numbers)



Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. The graph shows a wavescan plot across the range 220 nm to 750 nm with cursors denoting 230, 260, 280 and (if background correction selected) 320 nm.
- 4) Toggle on/off to display the data table only in the print-out.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , **OR** wait.

4.1.4 Dye incorporation for dsDNA, ssDNA, RNA and Oligonucleotides

The dye incorporation methods are similar to the dsDNA, ssDNA, RNA and Oligonucleotide methods. This section describes the specific features concerning the dye incorporation. For general information please follow the detailed instructions under **Analysis of dsDNA, ssDNA and RNA** or **Analysis of Oligonucleotides**.

To determine the dye incorporation rate, the absorbance reading at the wavelength reported for maximum absorbance of the fluorescence dye is used. For further details please refer to 11.2 Nucleic acid fluorescent dye incorporation.

The procedure is as follows:

Parameter Screen

LabelGuard Applications

dsDNA-Dye - Parameters

Lid Factor	Units
10	ng/μl
Dilution Factor	Factor
1.000	50.0
Background	Dye-Type
On	Alexa Fluor 350
OK	Cancel

Cuvette Applications

dsDNA-Dye - Parameters

Pathlength	Units
10 mm	ng/μl
Dilution Factor	Factor
1.000	50.0
Background	Dye-Type
On	Alexa Fluor 350
OK	Cancel

Results Screen

dsDNA-Dye

A230	0.076 A	Sample	1
A260	0.119 A	Concentration	54.22 ng/μl
A280	0.046 A	FOI (Cy3)	52.9
A320	0.009 A	Dye Concentration	8.80 pmol/μl
A550	0.132 A		
A650	0.066 A		
ADye(550)	0.132 A		
A260/A280	2.587		
A260/A230	1.566		

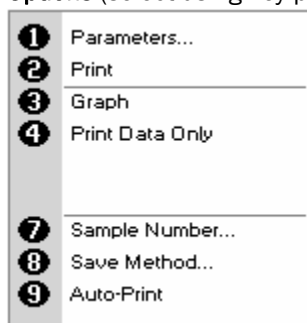
Parameter Screen

- Step 1** Press 1 for **LabelGuard** OR 2 for **Cuvette** folder.
- Step 2** Press 1 to select **Nucleic Acids** folder.
- Step 3** Press 5, 6, 7 or 8 to select one of the dye incorporation methods.
- Step 4** Using the LabelGuard™ Applications select the **Lid Factor** as described under 3.2. Using Cuvette Applications select **Pathlength** using the left and right arrows. Options are 5 mm or 10 mm.
- Step 5** Select **Dilution Factor**, **Units** and **Factor** as described under 4.1.2, 4.1.3.
- Step 6** **Background** correction at 320 nm is recommended to be switched on.
- Step 7** Select the appropriate **Dye Type**. 10 different AlexaFluors, 4 Cy-Dyes, 6 Oyster-Dyes and Texas Red are programmed with their corresponding maximum absorbance wavelength, dye-dependent correction factor at 260 nm and dye-dependent extinction coefficient. For further details please refer to 11.2 Nucleic acid fluorescent dye incorporation.

Results Screen

- Step 8** Insert the reference sample. Press **Blank** Key. This will be used for all subsequent samples until changed.
- Step 9** Insert sample and press . This measures at the selected wavelengths and displays the results. The sample and dye concentration, the FOI and the ratio of A260/A280 and A260/A230 are calculated (corrected by the background if selected).
- Step 10** If the absorbance value of the sample is not between 0.02 and 1.7 a **Warning message** and **Instruction** will be displayed in the top left corner of the result screen. Please refer to 3.2 Software instructions for further information.
- Step 11** Repeat for all samples.
- Step 12** Press **Options** to display available Options which are described below.
- Step 13** Press and confirm with to return to the **Nucleic Acids** folder.

Options (select using key pad numbers)



Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. The graph shows a wavescan plot across the range 220 nm to 750 nm with cursors denoting 230, 260, 280 and (if background correction selected) 320 nm.
- 4) Toggle on/off to display the data table only in the print-out.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , OR wait.

4.2 Protein Determination

4.2.1 General Information

Protein determination at 280 nm (LabelGuard Applications and Cuvette Applications)

- Protein can be determined in the near UV at 280 nm due to absorption by tyrosine, tryptophan and phenylalanine amino acids; Abs 280 varies greatly for different proteins due to their amino acid content, and consequently the specific absorption value for a particular protein must be determined.
- The presence of nucleic acid in the protein solution can have a significant effect due to strong nucleotide absorbance at 280 nm. This can be compensated by measuring Abs 260, and applying the equation of Christian and Warburg for the protein crystalline yeast enolase (Biochemische Zeitung 310, 384 (1941)):
$$\text{Protein (mg/ml)} = 1.55 * \text{Abs 280} - 0.76 * \text{Abs 260}$$

or, Protein conc. = (A280 Factor * Abs 280) - (A260 Factor * Abs 260)
with A260 Factor = Correction factor for absorbance reading at 260nm
and A280 Factor = Correction factor for absorbance reading at 280nm
- This equation can be applied to other proteins if the corresponding factors are known (please note that the factor used by the NanoPhotometer is the reciprocal value of the extinction coefficient from a protein). The instrument can determine protein concentration at 280 nm and uses the above equation as default; the factors can be changed, and the use of background correction at 320 nm is optional.
- To customize the equation for a particular protein, the absorbance values at 260 and 280 nm should be determined at known protein concentrations to generate simple simultaneous equations; solving these provides the two coefficients. In cases where A260 Factor is found to be negative, it should be set to zero since it means there is no contribution to the protein concentration due to absorbance at 260 nm.
- Set A260 Factor = 0.00 for direct λ 280 UV protein measurement; A280 Factor is based on the extinction coefficient of the protein [molecular weight/molar extinction coefficient]. If BSA (bovine serum albumin) is an acceptable standard, setting A280 Factor = 1.588 will give linear results from 0.1 to 160 mg/ml protein.
$$\text{Protein (mg/ml)} = 1.588 * \text{Abs 280}$$
- Rapid measurements such as this at 280 nm are particularly useful after isolation of proteins and peptides from mixtures using spin and HiTrap columns by centrifuge and gravity, respectively.

Protein determination at 280 nm and degree of labelling (LabelGuard Applications and Cuvette Applications)

- To determine the degree of labelling, the absorbance reading at the wavelength reported for maximum absorbance of the fluorescence dye is used. The corresponding extinction coefficient of the dye is used in the Lambert-Beer Law to determine the dye concentration ($c = A / (e * d)$). Absorbance values and extinction coefficients are used to calculate the dye per protein ratio. For further details please refer to 11.3 Protein fluorescent dye incorporation.

Colorimetric Bradford, Biuret, BCA and Lowry protein determination (Cuvette Applications)

- The **Bradford** method depends on quantifying the binding of a dye, Coomassie Brilliant Blue, to an unknown protein and comparing this binding to that of different, known concentrations of a standard protein at 595 nm; this is usually BSA (bovine serum albumin).
- The **Biuret** method depends on reaction between cupric ions and peptide bonds in an alkali solution, resulting in the formation of a complex absorbing at 546 nm.
- The **BCA** method also depends on reaction between cupric ions and peptide bonds, but in addition combines this reaction with the detection of cuprous ions using bicinchoninic acid (BCA), giving an absorbance maximum at 562 nm. The BCA process is less sensitive to the presence of detergents used to break down cell walls.
- The **Lowry** method is based on the Biuret reaction. Under alkaline conditions the divalent copper ion forms a complex with peptide bonds in which it is reduced to a monovalent ion. Monovalent copper ion and the radical groups of tyrosine, tryptophan, and cysteine react with Folin reagent to produce an unstable product that becomes reduced to molybdenum/tungsten blue. Bound reagent changes colour from yellow to blue. This binding is compared with those derived from a standard protein at 750 nm; this is usually BSA (bovine serum albumin).
- Detailed protocols are supplied with these assay kits, and must be closely followed to ensure accurate results are obtained.
- A linear regression analysis of the calibration standard data points is calculated; the result, together with the correlation coefficient, can be printed out. A correlation coefficient of between 0.95 and 1.00 indicates a good straight line.

4.2.2 Protein UV Method

The procedure is as follows:

Parameter Screen

LabelGuard Applications

Protein-UV - Parameters	
Lid Factor 10	A260 Factor 0.760
Dilution Factor 1.000	A280 Factor 1.550
Background On	Units mg/ml
OK	Cancel

Cuvette Applications

Protein UV - Parameters	
Pathlength 10 mm	A260 Factor 0.760
Dilution Factor 1.000	A280 Factor 1.550
Background On	Units mg/ml
OK	Cancel

Results Screen

Protein-UV	
A260 0.000 A	Sample Reference
A280 0.000 A	Result 0.000
A320 0.000 A	Units mg/ml

Parameter Screen

- Step 1** Press 1 for **LabelGuard** OR 2 for **Cuvette** folder.
- Step 2** Press 2 to select **Protein** folder.
- Step 3** Press 1 to select **Protein UV** mode.
- Step 4** Using LabelGuard Applications select the **Lid Factor** as described under 3.2. A minimum of **2 µl sample** volume is recommended. Using Cuvette Applications select **Pathlength** using the left and right arrows. Options are 5 mm or 10 mm.
- Step 5** Enter the **Dilution Factor** using the keypad numbers. Range 1.00 to 9999. Use the C button to backspace and clear the last digit entered. **OR** Press **Options** to enter the dilution factor screen. Enter the volume of the sample using the keypad numbers. Range 0.01 to 9999. Enter the volume of the diluent using the keypad numbers. Range 0.01 to 9999. Press $\diamond$ to calculate the dilution factor and return to the Parameters screen **OR** Press **Cancel** ∇ to cancel the selections and return to the Parameters screen.
- Step 6** Select whether the **Background** correction at 320 nm is used or not with the left and right arrows. It is recommended to switch on the **Background** correction.
- Step 7** Enter **A260 Factor** using the keypad numbers. Default value is 0.76. Range is 1.00 to 9999.
- Step 8** Enter **A280 Factor** using the keypad numbers. Default value is 1.55, range is 1.00 to 9999.
- Step 9** Select the **Units** of measurement using the left and right arrows. Options: mg/ml, µg/ml, ng/µl and µg/µl.
- Step 10** Press **OK** $\diamond$ to enter the Results screen **OR** **Cancel** ∇ to return to the **Protein** folder.

Results Screen

- Step 11** Insert the reference sample. Press **Blank** Key. This will be used for all subsequent samples until changed.
- Step 12** Insert sample and press $\diamond$. This measures at both 260 and 280 nm wavelengths and displays the result. Protein concentration is calculated (corrected by background wavelength value if selected).
- Step 13** If the absorbance value of the sample is not between 0.02 and 1.7 a **"Warning message"** and **"Instruction"** will be displayed in the top left corner of the result screen. Please refer to 3.2 Software instructions for further information.
- Step 14** Repeat for all samples.
- Step 15** Press **Options** to display available Options which are described below.
- Step 16** Press ∇ and confirm with $\diamond$ to return to the **Protein** folder.

Step 17

Options (select using key pad numbers)

1	Parameters...
2	Print
3	Graph
4	Print Data Only
7	Sample Number...
8	Save Method...
9	Auto-Print

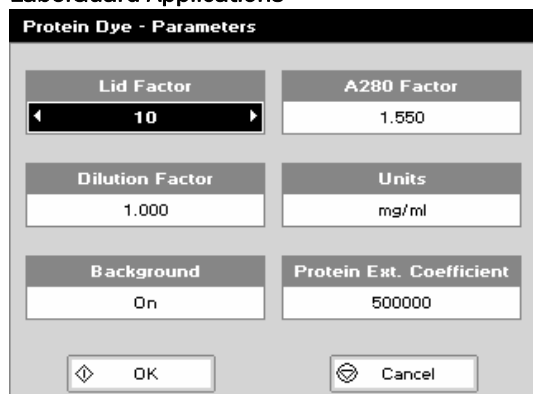
Options (select using key pad numbers)

- 1) Return to parameters screen.
 - 2) Print result via selected method.
 - 3) Toggle graph on/off. The graph shows a wavescan plot across the scan range 210 nm to 760 nm with cursors denoting 260 nm and 280 nm.
 - 4) Toggle on/off to display the data table only in the print-out.
Default setting prints always the graph plus the data.
 - 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
 - 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
 - 9) Auto-print – toggles auto-print on/off.
- Exit options by pressing , or wait.

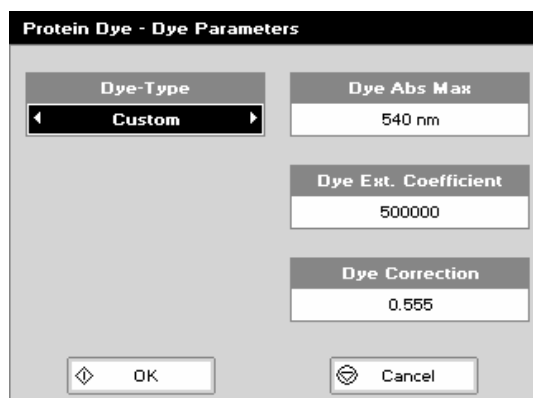
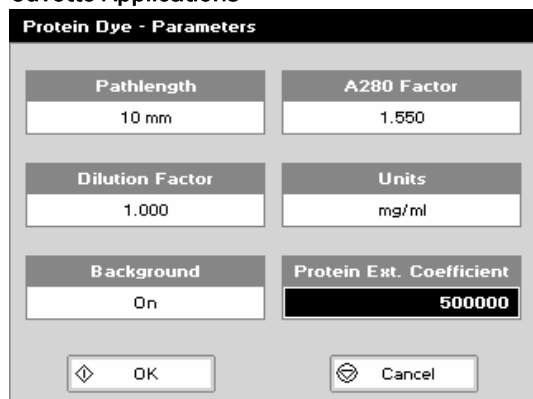
4.2.3 Protein UV Dye Method

The procedure is as follows:

Parameter Screen LabelGuard Applications



Cuvette Applications



Parameter Screen

- Step 1** Press 1 for **LabelGuard** OR 2 for **Cuvette** folder.
- Step 2** Press 2 to select **Protein** folder.
- Step 3** Press 2 to select **Protein dye** mode.
- Step 4** Using LabelGuard Applications select the **Lid Factor** as described under 3.2. A minimum of **2 µl sample** volume is recommended. Using Cuvette Applications select **Pathlength** using the left and right arrows. Options are 5 mm or 10 mm.
- Step 5** Enter the **Dilution Factor** using the keypad numbers. Range 1.00 to 9999. Use the C button to backspace and clear the last digit entered. **OR** Press **Options** to enter the dilution factor screen. Enter the volume of the sample using the keypad numbers. Range 0.01 to 9999. Enter the volume of the diluent using the keypad numbers. Range 0.01 to 9999. Press  to calculate the dilution factor and return to the Parameters screen **OR** Press Cancel  to cancel the selections and return to the Parameters screen.
- Step 6** Select whether the **Background** correction at 320 nm is used or not with the left and right arrows. It is recommended to switch on the **Background** correction.
- Step 7** Enter **A280 Factor** using the keypad numbers. Default value is 1.55. Range is 1.00 to 9999.
- Step 8** Select the **Units** of measurement using the left and right arrows. Options: mg/ml, µg/ml, ng/µl and µg/µl.
- Step 9** Enter the protein-dependent extinction coefficient. Range is 10000 to 9999999.
- Step 10** Press OK  to store the chosen parameters and to enter the next screen **OR** Cancel  to return to the **Protein** folder.
- Step 11** Select the appropriate Dye Type. 4 different AlexaFluors, 2 Cy-Dyes, 2 DyLight Dyes, FITC, Pacific Blue, r-PE and Texas Red are programmed with their corresponding maximum absorbance wavelength, dye-dependent extinction coefficient and dye-dependent correction factor at 280 nm.
- Step 12** If using **Custom Dye** maximum absorbance wavelength of the custom dye, dye-dependent extinction coefficient and dye-dependent correction factor at 280 nm have to be entered. For further details please refer to 11.3 Protein fluorescent dye incorporation.
Range are:
Dye Abs Max: 190 nm to 1100 nm
Dye Ext. Coefficient: 10000 to 9999999
Dye Correction: 0.001 to 0.999

Results Screen

Protein Dye	
A260	0.125 A
A280	0.163 A
A320	0.009 A
ADye(654)	0.184 A
Sample 1	
Protein Concentration 1.12 mg/ml	
Dye Concentration 7.36 pmol/μl	
Degree of Labelling 0.99	

Options (select using key pad numbers)

1	Parameters...
2	Print
3	Graph
4	Print Data Only
7	Sample Number...
8	Save Method...
9	Auto-Print

Results Screen

- Step 13** Insert the reference sample. Press **Blank** Key. This will be used for all subsequent samples until changed.
- Step 14** Insert sample and press . This measures at 260nm, 280nm, 320nm and the dye specific wavelength and displays the result. Protein concentration (corrected by background wavelength value if selected), dye concentration and degree of labelling is calculated.
- Step 15** If the absorbance value of the sample is not between 0.02 and 1.7 a **"Warning message"** and **"Instruction"** will be displayed in the top left corner of the result screen. Please refer to 3.2 Software instructions for further information.
- Step 16** Repeat for all samples.
- Step 17** Press **Options** to display available Options which are described below.
- Step 18** Press and confirm with to return to the **Protein** folder.

Options (select using key pad numbers)

- 1) Return to parameters screen.
 - 2) Print result via selected method.
 - 3) Toggle graph on/off. The graph shows a wavescan plot across the scan range 210 nm to 760 nm with cursors denoting 260 nm and 280 nm.
 - 4) Toggle on/off to display the data table only in the print-out.
Default setting prints always the graph plus the data.
 - 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
 - 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
 - 9) Auto-print – toggles auto-print on/off.
- Exit options by pressing , or wait.

4.2.4 BCA Assay

The colorimetric BCA assay is not recommended with the LabelGuard™ Microliter Cell. Please use Cuvette Applications. The procedure is as follows:

Parameter Screen

BCA - Parameters

Wavelength	Pathlength
562 nm	10 mm

Standards: 6

Units: µg/ml

Next Cancel

BCA - Parameters

Units

Units: µg/ml

DP: Auto

OK Cancel

BCA - Parameters

Curve Fit: Regression

Calibration: Standards

Replicates: Off

Next Cancel

Standards Screen

BCA - Standards

Std. 1	Std. 4
0.200	0.800
Std. 2	Std. 5
0.400	1.000
Std. 3	Std. 6
0.600	1.400

Next Back

Parameter Screen

- Step 1** Press 2 to select **Cuvette** folder.
- Step 2** Press 2 to select **Protein** folder.
- Step 3** Press 2 to select **BCA** mode.
- Step 4** The default **Wavelength** setting is 562 nm.
- Step 5** Enter the number of **Standard** concentration points (1-9) to be used in the curve using the keypad numbers or left and right arrows.
- Step 6** Select **Pathlength** using the left and right arrows. Options are 5 or 10 mm.

- Step 7** **Units:** The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows (µg/ml, µg/µl, pmol/µl, mg/dl, mmol/l, µmol/l, g/l, mg/l, µg/l, U/l, %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2. Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press **OK** to store the chosen parameters **OR** **Cancel**.

- Step 8** Press **Next** to enter the next screen.

- Step 9** Select the **Calibration** mode, either standards (measure prepared standards), manual (keypad data entry) or new standards (using a saved method previous values are blanked, new standard can be measured).

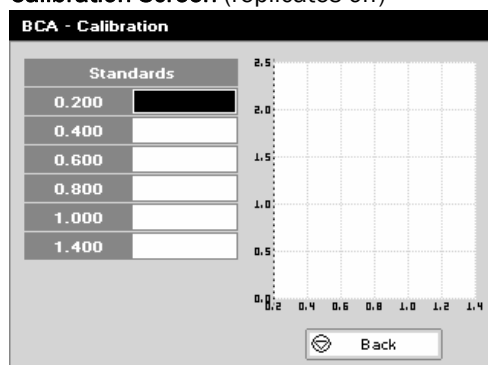
- Step 10** (if standards selected) Select the number of **Replicates** using the left and right arrows. This determines the number of standards to be measured and averaged at each standard concentration point. Can be OFF (1), 2 or 3.

- Step 11** Press **Next** to enter the Standards screen **OR** Press **Cancel** to cancel selections and return to the **Protein** folder.

Standards Screen

- Step 12** Enter the concentration values by using the keypad numbers and the up and down arrows to move between the different standard boxes. Range 0.001 to 9999. C button backspaces and clears the last digit entered.
- Step 13** Press **Next** to enter the Calibration screen. If there are duplicate or non-monotonic (increasing) entries the unit will beep and highlight the incorrect entry. **OR** Press **Back** to return to the Parameter screen.

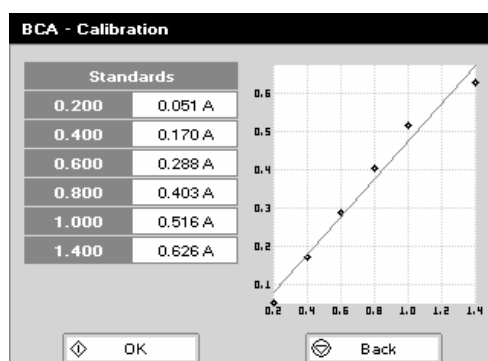
Calibration Screen (replicates off)



Calibration Screen (replicates off)

Step 14 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

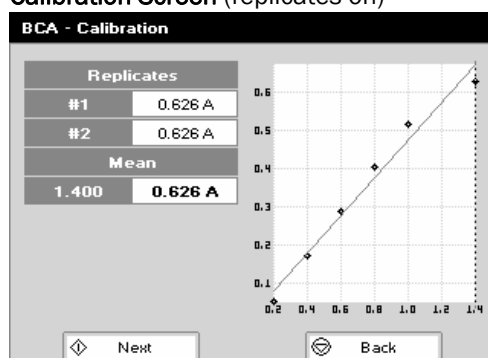
Step 15 Insert the standard (use C to clear previously stored results before measuring). Press to measure the standard and store the result.



Step 16 Repeat for all standards. A graph will display the results and the fitted curve as the measurements are made. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 17 When all standards are measured press to accept the calibration and go to the Results screen (see below) **OR** Press Back to cancel selections and return to the Standards screen.

Calibration Screen (replicates on)



Calibration Screen (replicates on)

Step 18 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

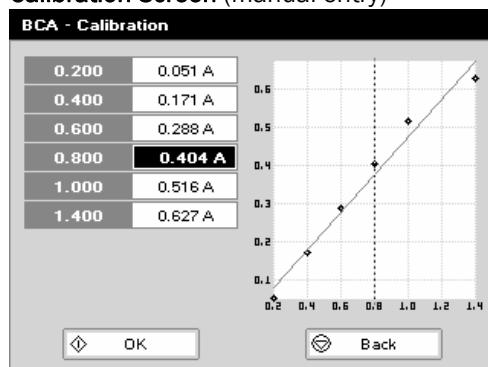
Step 19 Press to display the replicate entry boxes. Use C to clear previously stored results before measuring.

Step 20 Insert the standard and press to measure the standard and store the result.

Step 21 Repeat for all replicates and standards. A graph will display the results and the fitted curve as the measurements are made. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 22 Press to accept the calibration and go to the Results screen (see below) **OR** Press Back to return to the Standards screen.

Calibration Screen (manual entry)



Calibration Screen (manual entry)

Shows previously entered calibration values and allows values to be entered via the keypad.

Step 23 The highlighted box can be edited in order to enter an absorbance value corresponding to a given concentration value using the keypad numbers. Range 0.001 to 9999. Use C to backspace and clear the last digit entered and the up and down arrows to move between boxes.

Step 24 Press OK to accept the calibration and go to the Results screen (see below) OR Press Back to return to the Standards screen.

Results screen

The Results screen displays the following information:

BCA	
Wavelength 562 nm	Sample 1
Absorbance 0.288 A	Units µg/ml
Curve Fit Regression	Concentration 0.622
Standard Pathlength 10 mm	Sample Pathlength 10 mm

Results screen

Step 25 Insert the reference sample and press **Blank** key. This will be used for all subsequent samples until changed.

Step 26 Insert the sample and press . The concentration of the sample is taken and displayed.

Step 27 Repeat for all samples.

Step 28 Press **Options** to display available Options which are described below.

Step 29 Press and confirm with to return to the **Protein** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

The Options menu displays the following numbered options:

- 1 Parameters...
- 2 Print
- 3 Graph
- 4 Edit Sample Pathlength
- 7 Sample Number...
- 8 Save Method...
- 9 Auto-Print

Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. Displays the calibration graph, cursors give values for last measured sample.
- 4) Possibility to edit the sample pathlength.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

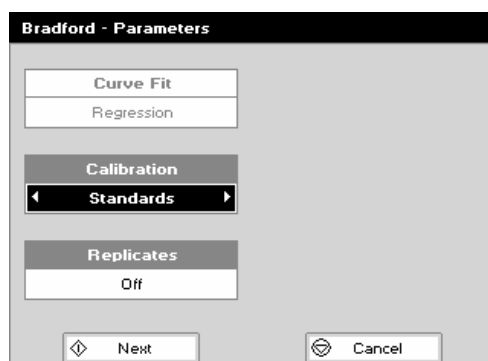
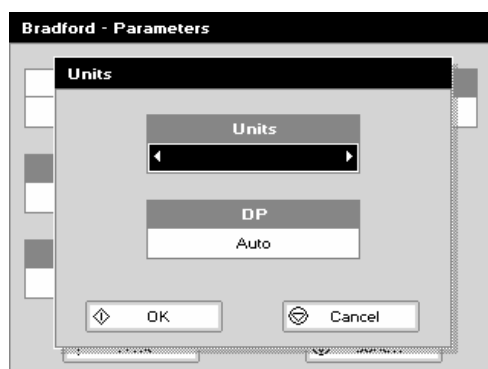
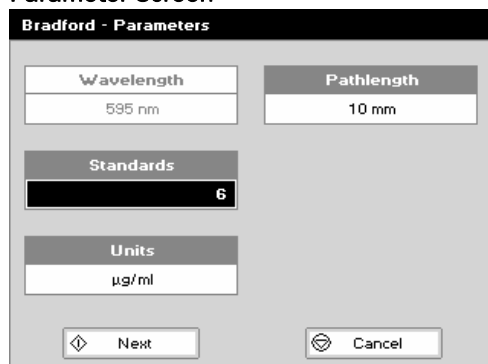
Exit options by pressing , or wait.

4.2.5 Bradford Assay

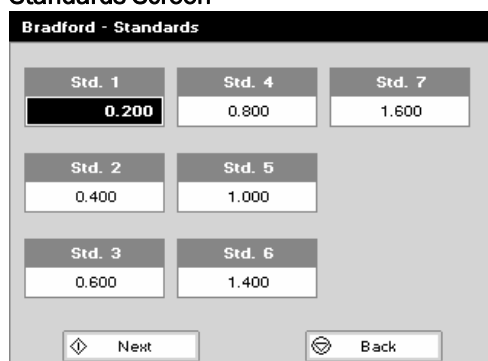
The colorimetric Bradford assay is not recommended with the LabelGuard™ Microliter Cell. Please use Cuvette Applications.

The procedure is as follows:

Parameter Screen



Standards Screen



Parameter Screen

- Step 1** Press 2 to select **Cuvette** folder.
- Step 2** Press 2 to select **Protein** folder.
- Step 3** Press 3 to select **Bradford** mode.
- Step 4** The default **Wavelength** setting is 595 nm.
- Step 5** Enter the number of **Standard** concentration points (1-9) to be used in the curve using the keypad numbers or left and right arrows.
- Step 6** Select **Pathlength** using the left and right arrows. Options are 5 or 10 mm.

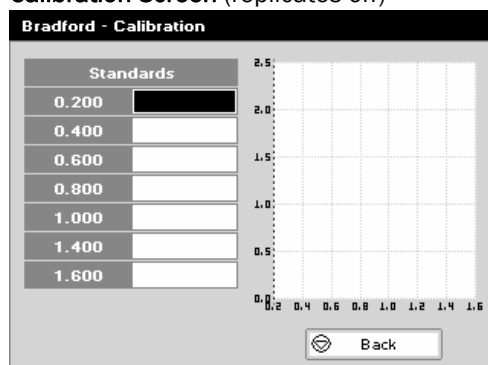
- Step 7** **Units:** The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows (µg/ml, µg/µl, pmol/µl, mg/dl, mmol/l, µmol/l, g/l, mg/l, µg/l, U/l, %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2. Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press OK  to store the chosen parameters **OR** Cancel . Press Next  to enter the next screen.

- Step 8** Select the **Calibration** mode, either standards (measure prepared standards), manual (keypad data entry) or new standards (using a saved method previous values are blanked, new standard can be measured).
- Step 9** (if standards selected) Select the number of **Replicates** using the left and right arrows. This determines the number of standards to be measured and averaged at each standard concentration point. Can be OFF (1), 2 or 3.
- Step 10** Press Next  to enter the Standards screen **OR** Press Cancel  to cancel selections and return to the **Protein** folder.

Standards Screen

- Step 11** Enter the concentration values by using the keypad numbers and the up and down arrows to move between the different standard boxes. Range 0.001 to 9999. C button backspaces and clears the last digit entered.
- Step 12** Press Next  to enter the Calibration screen. If there are duplicate or non-monotonic (increasing) entries the unit will beep and highlight the incorrect entry. **OR** Press Back  to return to the Parameter screen.

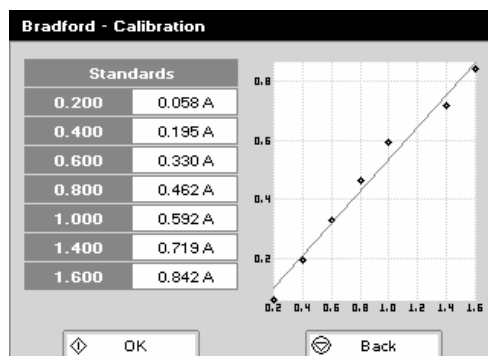
Calibration Screen (replicates off)



Calibration Screen (replicates off)

Step 13 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

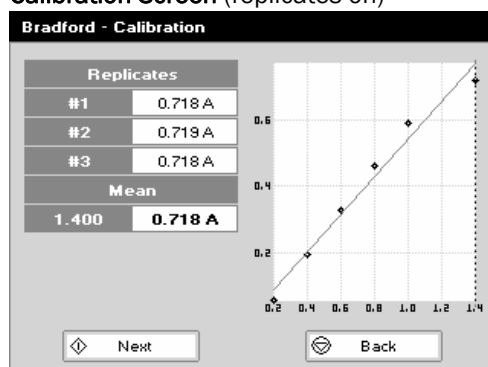
Step 14 Insert the standard (use C to clear previously stored results before measuring). Press to measure the standard and store the result.



Step 15 Repeat for all standards. A graph will display the results and the fitted curve as the measurements are made. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 16 When all standards are measured press to accept the calibration and go to the Results screen (see below) **OR** Press Back to cancel selections and return to the Standards screen.

Calibration Screen (replicates on)



Calibration Screen (replicates on)

Step 17 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

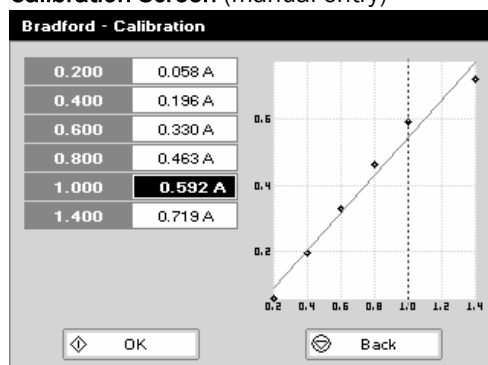
Step 18 Press to display the replicate entry boxes. Use C to clear previously stored results before measuring.

Step 19 Insert the standard and press to measure the standard and store the result.

Step 20 Repeat for all replicates and standards. A graph will display the results and the fitted curve as the measurements are made. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 21 Press to accept the calibration and go to the Results screen (see below) **OR** Press Back to return to the Standards screen.

Calibration Screen (manual entry)



Calibration Screen (manual entry)

Shows previously entered calibration values and allows values to be entered via the keypad.

Step 22 The highlighted box can be edited in order to enter an absorbance value corresponding to a given concentration value using the keypad numbers. Range 0.001 to 9999. Use C to backspace and clear the last digit entered and the up and down arrows to move between boxes.

Step 23 Press OK to accept the calibration and go to the Results screen (see below) OR Press Back to return to the Standards screen.

Results screen

The results screen displays the following information:

Bradford	
Wavelength	Sample 1
595 nm	
Absorbance	Units µg/ml
0.331 A	
Curve Fit	Concentration
Regression	0.624
Standard Pathlength	Sample Pathlength
10 mm	10 mm

Results screen

Step 24 Insert the reference sample and press **Blank** key. This will be used for all subsequent samples until changed.

Step 25 Insert the sample and press . The concentration of the sample is taken and displayed.

Step 26 Repeat for all samples.

Step 27 Press **Options** to display available Options which are described below.

Step 28 Press and confirm with to return to the **Protein** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

The options menu displays the following settings:

- 1 Parameters...
- 2 Print
- 3 Graph
- 4 Edit Sample Pathlength
- 7 Sample Number...
- 8 Save Method...
- 9 Auto-Print

Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. Displays the calibration graph, cursors give values for last measured sample.
- 4) Possibility to edit the sample pathlength.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

4.2.6 Lowry Assay

The colorimetric Lowry assay is not recommended with the LabelGuard™ Microliter Cell. Please use Cuvette Applications. The procedure is as follows:

Parameter Screen

Lowry - Parameters

Wavelength	750 nm	Pathlength	10 mm
Standards	6		
Units	µg/ml		

Next Cancel

Lowry - Parameters

Units

Units	µg/ml
DP	Auto

OK Cancel

Lowry - Parameters

Curve Fit	Regression
Calibration	Standards
Replicates	Off

Next Cancel

Standards Screen

Lowry - Standards

Std. 1	0.200	Std. 4	0.800
Std. 2	0.400	Std. 5	1.000
Std. 3	0.600	Std. 6	1.400

Next Back

Parameter Screen

- Step 1** Press 2 to select **Cuvette** folder.
Step 2 Press 2 to select **Protein** folder.
Step 3 Press 4 to select **Lowry** mode.
Step 4 The default **Wavelength** setting is 750 nm.
Step 5 Enter the number of **Standard** concentration points (1-9) to be used in the curve using the keypad numbers or left and right arrows.
Step 6 Select **Pathlength** using the left and right arrows. Options are 5 or 10 mm.

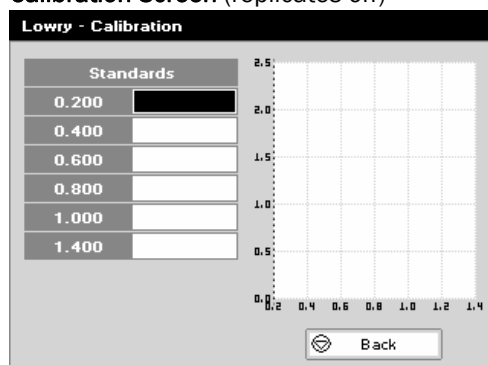
- Step 7 Units:** The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows (µg/ml, µg/µl, pmol/µl, mg/dl, mmol/l, µmol/l, g/l, mg/l, µg/l, U/l, %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2. Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press OK to store the chosen parameters **OR** Cancel. Press Next to enter the next screen.

- Step 8** Select the **Calibration** mode, either standards (measure prepared standards), manual (keypad data entry) or new standards (using a saved method previous values are blanked, new standard can be measured).
Step 9 (if standards selected) Select the number of **Replicates** using the left and right arrows. This determines the number of standards to be measured and averaged at each standard concentration point. Can be OFF (1), 2 or 3.
Step 10 Press Next to enter the Standards screen **OR** Press Cancel to cancel selections and return to the **Protein** folder.

Standards Screen

- Step 11** Enter the concentration values by using the keypad numbers and the up and down arrows to move between the different standard boxes. Range 0.001 to 9999. C button backspaces and clears the last digit entered.
Step 12 Press Next to enter the Calibration screen. If there are duplicate or non-monotonic (increasing) entries the unit will beep and highlight the incorrect entry. **OR** Press Back to return to the Parameter screen.

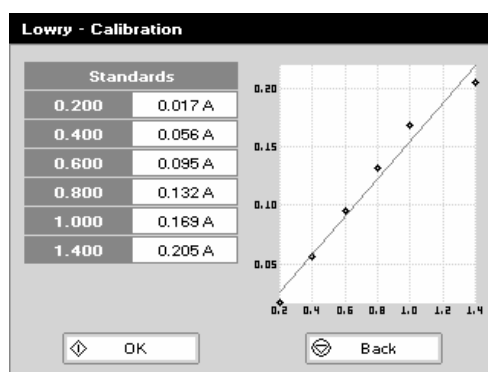
Calibration Screen (replicates off)



Calibration Screen (replicates off)

Step 13 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

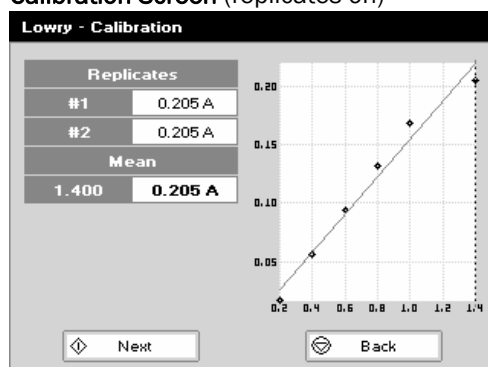
Step 14 Insert the standard (use C to clear previously stored results before measuring). Press to measure the standard and store the result.



Step 15 Repeat for all standards. A graph will display the results and the fitted curve as the measurements are made. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 16 When all standards are measured press to accept the calibration and go to the Results screen (see below) **OR** Press Back to cancel selections and return to the Standards screen.

Calibration Screen (replicates on)



Calibration Screen (replicates on)

Step 17 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

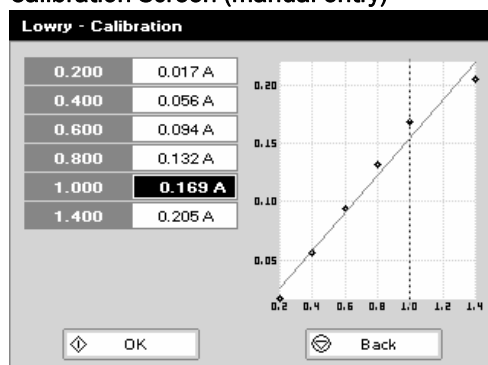
Step 18 Press to display the replicate entry boxes. Use C to clear previously stored results before measuring.

Step 19 Insert the standard and press to measure the standard and store the result.

Step 20 Repeat for all replicates and standards. A graph will display the results and the fitted curve as the measurements are input. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 21 Press to accept the calibration and go to the Results screen (see below) **OR** Press Back to return to the Standards screen.

Calibration Screen (manual entry)



Calibration Screen (manual entry)

Shows previously entered calibration values and allows values to be entered via the keypad.

Step 22 The highlighted box can be edited in order to enter an absorbance value corresponding to a given concentration value using the keypad numbers. Range 0.001 to 9999. Use C to backspace and clear the last digit entered and the up and down arrows to move between boxes.

Step 23 Press OK to accept the calibration and go to the Results screen (see below) OR Press Back to return to the Standards screen.

Results screen

The results screen displays the following parameters for Sample 1:

Wavelength	750 nm	Sample	1
Absorbance	0.095 A	Units	µg/ml
Curve Fit	Regression	Concentration	0.627
Standard Pathlength	10 mm	Sample Pathlength	10 mm

Results screen

Step 24 Insert the reference sample and press **Blank** key. This will be used for all subsequent samples until changed.

Step 25 Insert the sample and press . The concentration of the sample is taken and displayed.

Step 26 Repeat for all samples.

Step 27 Press **Options** to display available Options which are described below Press and confirm with to return to the **Protein** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

The options menu lists the following options:

- 1 Parameters...
- 2 Print
- 3 Graph
- 4 Edit Sample Pathlength
- 7 Sample Number...
- 8 Save Method...
- 9 Auto-Print

Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. Displays the calibration graph, cursors give values for last measured sample.
- 4) Possibility to edit the sample pathlength.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

4.2.7 Biuret Assay

The colorimetric Biuret assay is not recommended with the LabelGuard™ Microliter Cell. Please use Cuvette Applications. The procedure is as follows:

Parameter Screen

The screen displays the following parameters:

Wavelength	Pathlength
546 nm	10 mm

Standards: 6

Units: µg/ml

Buttons: Next, Cancel

Parameter Screen

- Step 1** Press 2 to select **Cuvette** folder.
- Step 2** Press 2 to select **Protein** folder.
- Step 3** Press 5 to select **Biuret** mode.
- Step 4** The default **Wavelength** setting is 546 nm.
- Step 5** Enter the number of **Standard** concentration points (1-9) to be used in the curve using the keypad numbers or left and right arrows.
- Step 6** Select **Pathlength** using the left and right arrows. Options are 5 mm or 10 mm.

The screen displays the following parameters:

Units: µg/µl

DP: Auto

Buttons: OK, Cancel

- Step 7 Units:** The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows (µg/ml, µg/µl, pmol/µl, mg/dl, mmol/l, µmol/l, g/l, mg/l, µg/l, U/l, %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2. Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press OK to store the chosen parameters OR Cancel to return to the **Protein** folder.

The screen displays the following parameters:

Curve Fit: Regression

Calibration: Standards

Replicates: Off

Buttons: Next, Cancel

- Step 8** Select the **Calibration** mode, either standards (measure prepared standards), manual (keypad data entry) or new standards (using a saved method previous values are blanked, new standard can be measured).
- Step 9** (if standards selected) Select the number of **Replicates** using the left and right arrows. This determines the number of standards to be measured and averaged at each standard concentration point. Can be OFF (1), 2 or 3.
- Step 10** Press Next to enter the Standards screen OR Press Cancel to cancel selections and return to the **Protein** folder.

Standards Screen

The screen displays the following standard concentrations:

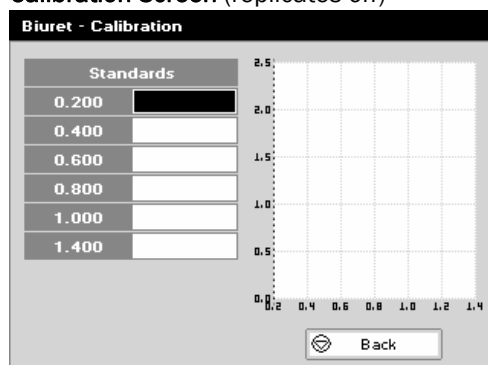
Std. 1	Std. 2	Std. 3	Std. 4	Std. 5	Std. 6
0.200	0.400	0.600	0.800	1.000	1.400

Buttons: Next, Back

Standards Screen

- Step 11** Enter the concentration values by using the keypad numbers and the up and down arrows to move between the different standard boxes. Range 0.001 to 9999. C button backspaces and clears the last digit entered.
- Step 12** Press Next to enter the Calibration screen. If there are duplicate or non-monotonic (increasing) entries the unit will beep and highlight the incorrect entry OR Press Back to return to the Parameter screen.

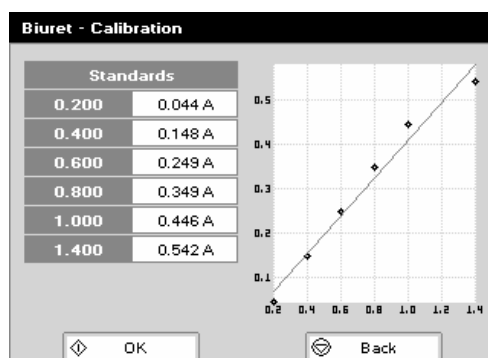
Calibration Screen (replicates off)



Calibration Screen (replicates off)

Step 13 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

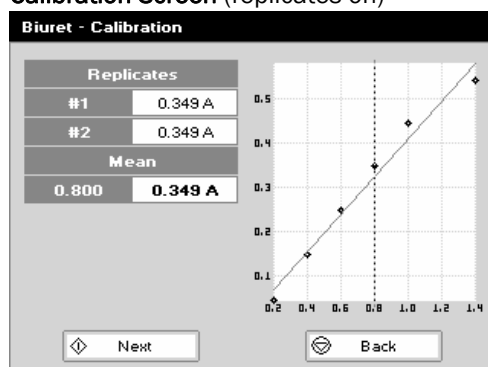
Step 14 Insert the standard (use C to clear previously stored results before measuring). Press $\uparrow$ to measure the standard and store the result.



Step 15 Repeat for all standards. A graph will display the results and the fitted curve as the measurements are made. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 16 When all standards are measured press $\uparrow$ to accept the calibration and go to the Results screen (see below) **OR** Press Back $\downarrow$ to cancel selections and return to the Standards screen.

Calibration Screen (replicates on)



Calibration Screen (replicates on)

Step 17 This shows the calibration values and allows standards to be measured. Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

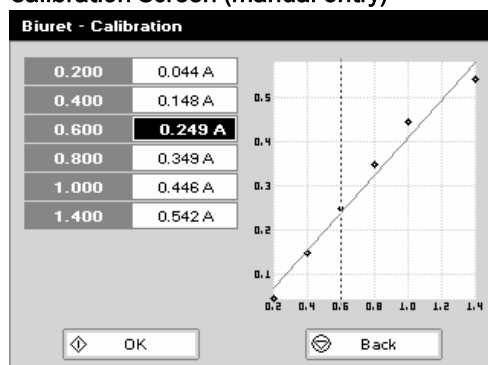
Step 18 Press $\uparrow$ to display the replicate entry boxes. Use C to clear previously stored results before measuring.

Step 19 Insert the standard and press $\uparrow$ to measure the standard and store the result.

Step 20 Repeat for all replicates and standards. A graph will display the results and the fitted curve as the measurements are input. Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 21 Press $\uparrow$ to accept the calibration and go to the Results screen (see below) **OR** Press Back $\downarrow$ to return to the Standards screen.

Calibration Screen (manual entry)



Calibration Screen (manual entry)

Shows previously entered calibration values and allows values to be entered via the keypad.

Step 22 The highlighted box can be edited in order to enter an absorbance value corresponding to a given concentration value using the keypad numbers. Range 0.001 to 9999. Use C to backspace and clear the last digit entered and the up and down arrows to move between boxes.

Step 23 Press OK to accept the calibration and go to the Results screen (see below) OR Press Back to return to the Standards screen.

Results screen

Biuret

Wavelength 546 nm	Sample 1
Absorbance 0.249 A	Units µg/µl
Curve Fit Regression	Concentration 0.622
Standard Pathlength 10 mm	Sample Pathlength 10 mm

Results screen

Step 24 Insert the reference sample and press **Blank** key. This will be used for all subsequent samples until changed.

Step 25 Insert the sample and press . The concentration of the sample is taken and displayed.

Step 26 Repeat for all samples.

Step 27 Press **Options** to display available Options which are described below.

Step 28 Press and confirm with to return to the **Protein** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

- 1 Parameters...
- 2 Print
- 3 Graph
- 4 Edit Sample Pathlength
- 7 Sample Number...
- 8 Save Method...
- 9 Auto-Print

Options (select using key pad numbers)

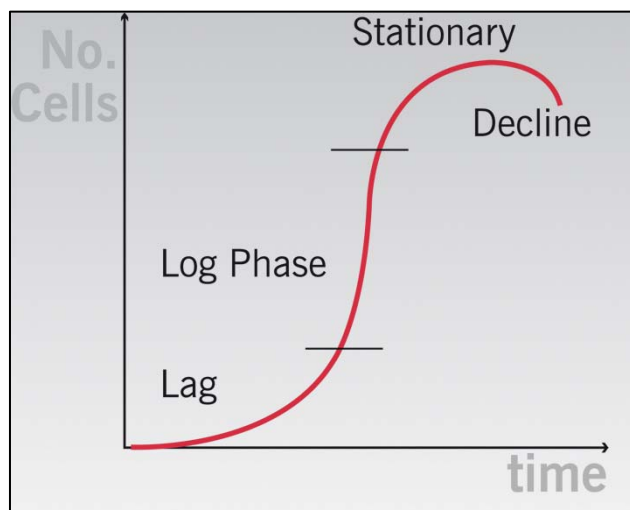
- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. Displays the calibration graph, cursors give values for last measured sample.
- 4) Possibility to edit the sample pathlength.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

4.3 Bacterial Cell Culture Measurement (OD600)

4.3.1 General Information

- The stage of growth of a bacterial culture needs to be monitored to ensure that the cells are harvested at the optimum point for the greatest density of live cells. An exemplary growth curve is given below. Cells should be harvested towards the end of the log phase. The optical density of the sample indicates when this point has been reached. This value varies dependent on the cells being grown. Routinely the cells are grown until the absorbance at 600 nm (known as OD 600) reaches approximately 0.4 prior to induction or harvesting. A linear relationship exists between cell number (density) and OD 600 up to approx. 0.6



- It is important to note that for turbid samples such as cell cultures, the absorbance measured is due to light scattering, and **not** the result of molecular absorption. The amount of scatter is affected by the optics of the system (distance between the cell holder and instrument exit slit, geometry of this slit and the monochromator optics). Different spectrophotometer types therefore give different responses for the same turbid sample; to compare results, they must be normalized using calibration curves.
- A calibration curve can be determined by comparing measured OD 600 to expected OD 600. Expected OD 600 is determined by counting cell number using an alternative technique (for example microscope slide method) and converting to OD 600 using the rule of thumb that $1 \text{ OD } 600 = 5 \times 10^8 \text{ cells/ml}$ for E. Coli.
- Additionally your NanoPhotometer™ is coming with a correction factor of 1 as default. To compare OD values between different spectrophotometer, you have to determine the constant deviation between the Absorbance values for the same sample within those instruments and use this factor within the setting "correction factor" of your NanoPhotometer™ Software.
- The use of 10 mm pathlength disposable cells is recommended for optical density measurements of cell culture solutions; to prevent the suspension settling too quickly and giving an OD that changes with time, glycerol should be added to the sample.
- The LabelGuard™ Microliter Cell is not recommended for optical density measurements of cell culture solutions.

4.3.2 Analysis of Bacterial Growth

The procedure is as follows:

Parameter Screen

OD 600 - Parameters

Wavelength	600 nm
Correction	1.000
Units	OD

OK Cancel

OD 600 - Parameters

Wavelength	600 nm	Factor	500
Correction	1.000	Multiplier	x 1000,000
Units	cells/ml		

OK Cancel

Results Screen

OD 600

Wavelength	950 nm	Sample	10
Absorbance	0.010 A	cells/ml	5.000
		Multiplier	x 1000,000

Options (select using key pad numbers)

- 1 Parameters...
- 2 Print
- 7 Sample Number...
- 8 Save Method...
- 9 Auto-Print

Parameter Screen

- Step 1** Press 2 to select **Cuvette Applications**.
- Step 2** Press 3 to select **OD 600**.
- Step 3** Select the **Wavelength**. Default value is 600 nm. Range is 200 nm to 950 nm.
- Step 4** Enter the **Correction** factor to compensate for different optical configurations between this and other instruments. Default value is 1.
- Step 5** Select the **Units**. Options are OD or cells/ml. If cells/ml is selected two further parameters are displayed.

- Step 6** (if cells/ml selected) Enter the **Factor** using the keypad numbers. Range 0.00 to 9999. C button backspaces and clears the last digit entered.
- Step 7** (if cells/ml selected) Select the **Multiplier** using the left and right arrows. Options are 1000 or 1,000,000. **Factor** and **Multiplier** define the conversion of the measured OD to the number of cells per millilitre (e.g.: 1 OD 600 = 5×10^8 cells/ml)
- Step 8** Press OK to enter the Results screen OR Press Cancel to cancel selections and return to the **Cuvette Applications** folder.

Results Screen

- Step 9** Insert the reference sample and press **Blank** key. This will be used for all subsequent samples until changed.
- Step 10** Insert the sample and press . The wavelength, absorbance and OD600 value is displayed.
- Step 11** Repeat for all samples.
- Step 12** Press **Options** to display available Options which are described below.
- Step 13** Press and confirm with to return to the **Cuvette Applications** folder. Query needs confirmation to avoid unintended escaping the application.

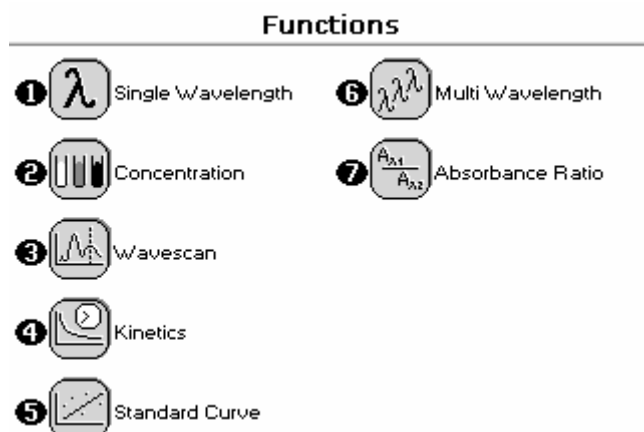
Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

5. FUNCTIONS

Survey of the available Functions:



Key pad number	Description
1	Absorbance or %T (transmission) at a single user defined wavelength.
2	Colorimetric assay at a single wavelength based on a simple Factor entered or calculated from a single standard.
3	Spectral plot between two user defined wavelengths. Range 200-950 nm, with user configurable peak finding function.
4	Kinetic colorimetric assay either rate or end value based.
5	Colorimetric assay at a single wavelength based on a user programmed curve.
6	Absorbance or %T (transmission) at up to 5 user defined wavelengths.
7	Ratio of absorbance values at two user specified wavelengths.

Options

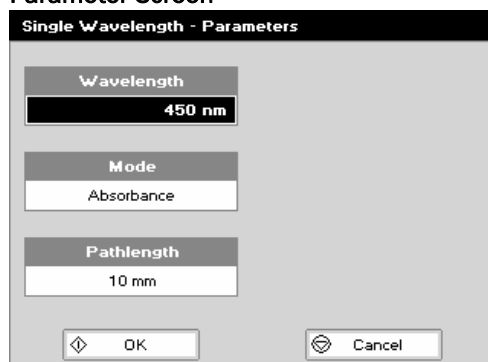
Within each function the user has the possibility to select various options that define the way results are treated. If not using a stored method, it is advisable to check that these Options have been appropriately set for your experiment when coming to the instrument. Note that setting the “History” parameter to on (see Preferences later) will cause the instrument to store it’s last settings. If the “History” parameter is turned off, all parameters and selections will return to their default settings when leaving that application. (Unless it has been saved as a method).

5.1 Single Wavelength – Abs and %T

This makes simple absorbance (A) and % transmission (%T) measurements on samples, measuring the amount of light that has passed through a sample relative to a reference (this can be air).

The procedure is as follows:

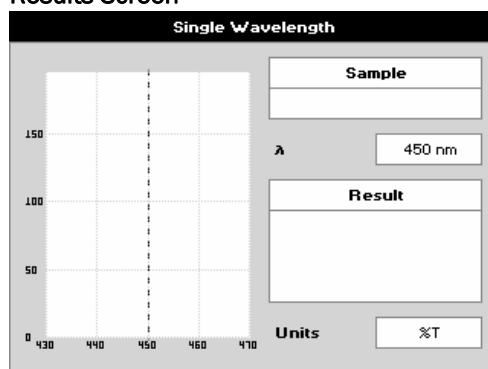
Parameter Screen



Parameter Screen

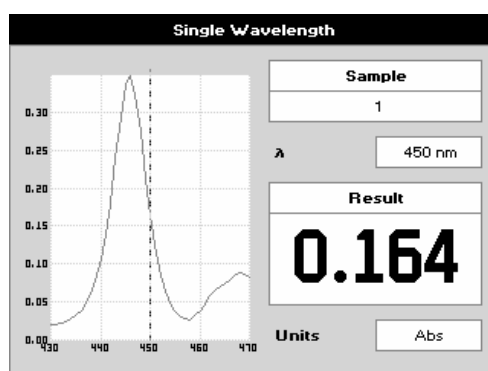
- Step 1** Press 3 to select **Functions**.
- Step 2** Press 1 to select **Single Wavelength**
- Step 3** Set **Wavelength** by using keypad numbers or left and right arrows.
- Step 4** Select the **Mode**, Absorbance or %Transmission, using the left and right arrows.
- Step 5** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.
- Step 6** To enter the results screen with the selected parameters press OK  OR cancel the selections and return to the **Functions** folder by pressing Cancel .

Results Screen



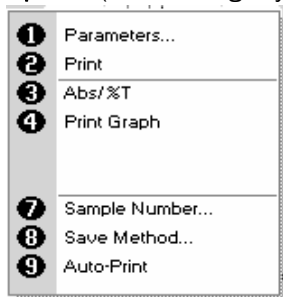
Results Screen

- Step 7** Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.
- Step 8** Insert sample and press .
- Step 9** Repeat for all samples.



- Step 10** The result at the selected wavelength is displayed on the screen.
- Step 11** Use the left and right arrows to move the cursor and display the value at the cursor position (+/- 15nm from set wavelength).
- Step 12** Press **Options** to display available Options which are described below.
- Step 13** Press Escape  and confirm with  to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application folder.

Options (select using key pad numbers)



Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle between Absorbance and %T mode.
- 4) Print graph – greyed out if no data are available.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

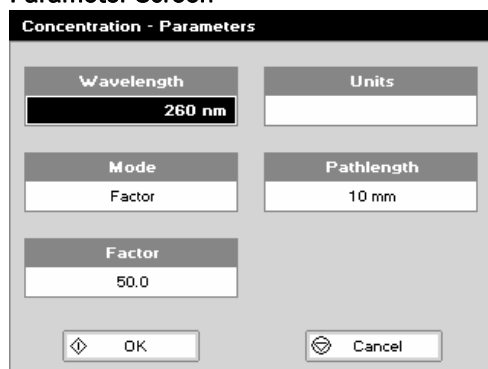
Exit options by pressing , or wait.

5.2 Concentration

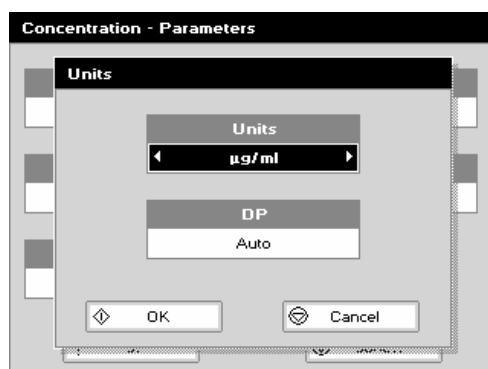
This makes simple concentration measurements on samples, by measuring the amount of light that has passed through a sample relative to a reference (this can be air). Concentration is obtained by multiplying the measured absorbance at a specific wavelength by a factor. The factor may be known in advance, or may be calculated by the instrument by measuring a standard of known concentration.

The procedure is as follows:

Parameter Screen

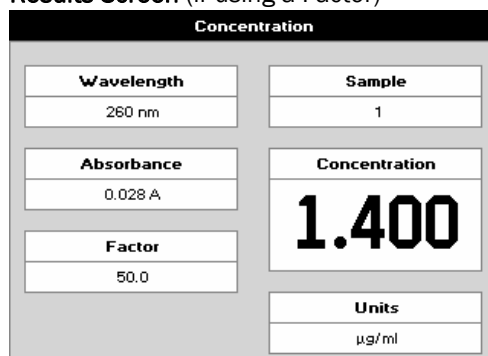


The screenshot shows the 'Concentration - Parameters' screen. It has a black header with the title. Below the header, there are four main sections: 'Wavelength' with a value of '260 nm', 'Units' (empty), 'Mode' with a value of 'Factor', and 'Pathlength' with a value of '10 mm'. At the bottom, there is a 'Factor' section with a value of '50.0'. At the very bottom, there are two buttons: 'OK' and 'Cancel'.



The screenshot shows a sub-screen for selecting units. It has a black header with the title 'Units'. Below the header, there is a 'Units' section with a dropdown menu showing 'µg/ml'. Below that, there is a 'DP' section with a value of 'Auto'. At the bottom, there are two buttons: 'OK' and 'Cancel'.

Results Screen (if using a Factor)



The screenshot shows the 'Concentration' results screen. It has a black header with the title. Below the header, there are four main sections: 'Wavelength' with a value of '260 nm', 'Sample' with a value of '1', 'Absorbance' with a value of '0.028 A', and 'Concentration' with a large value of '1.400'. At the bottom, there is a 'Factor' section with a value of '50.0' and a 'Units' section with a value of 'µg/ml'.

Parameter Screen

- Step 1** Press 3 to select **Functions**.
- Step 2** Press 2 to select **Concentration**
- Step 3** Set **Wavelength** by using keypad numbers or left and right arrows.
- Step 4** Select the **Mode**: Factor (user entered) or Standard (factor is calculated from a calibration sample), using the left and right arrows.
- Step 5** (if Factor is selected) Enter the **Factor** using the keypad numbers. Range 0.001 to 9999. Use the C button to delete the last digit entered.
- Step 6** (if Standard is selected) Enter the concentration using keypad numbers. Range 0.01-9999. Use the C button to delete the last digit entered.
- Step 7** **Units**: The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows (µg/ml, µg/µl, pmol/µl, mg/dl, mmol/l, µmol/l, g/l, mg/l, µg/l, U/l, %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2. Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press OK  to store the chosen parameters or Cancel .
- Step 8** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.
- Step 9** To enter the results screen with the selected parameters press  OR cancel the selections and return to the **Functions** folder by pressing Cancel .

Results Screen (if using a Factor)

- Step 10** Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.
- Step 11** Insert sample and press .

Results Screen (if using standard mode)

Options (select using key pad numbers)

Results Screen (if using standard mode)

- Step 12** Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.
- Step 13** Press to display the Run Standard screen.
- Step 14** Run the standard by pressing OR Press Cancel to return to the measure screen.

- Step 15** Insert the sample and press . The concentration of the sample is displayed. Results shown as — indicate the concentration is out of range.
- Step 16** Repeat for all samples.
- Step 17** Press **Options** to display available Options which are described below.
- Step 18** Press and confirm with to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggles on/off, displaying a graph of wavescan +/- 20 nm from selected wavelength.
- 4) Return to Run Standard screen.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

5.3 Wavescan

An absorption spectrum can be obtained from your instrument, enabling simple identification of peak height and position.

The procedure is as follows:

Parameter Screen

Wavescan - Parameters

Start Wavelength: 400 nm

End Wavelength: 500 nm

Mode: Absorbance

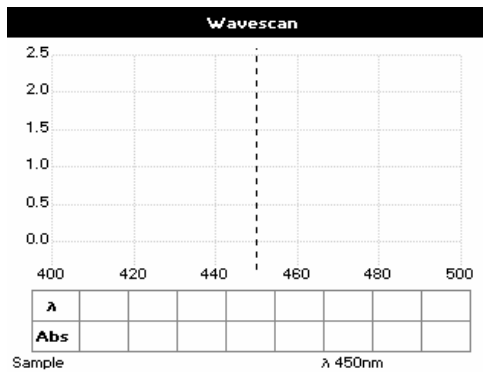
Pathlength: 10 nm

OK Cancel

Parameter Screen

- Step 1** Press 3 to select **Functions**.
- Step 2** Press 3 to select **Wavescan**.
- Step 3** Set **Start Wavelength** by using keypad numbers or left and right arrows.
- Step 4** Set **End Wavelength** by using keypad numbers or left and right arrows.
- Step 5** Select the **Mode**, **Absorbance** or **%Transmission**, using the left and right arrows.
- Step 6** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.
- Step 7** To enter the measurements screen with the selected parameters press OK  **OR** cancel the selections and return to the **Functions** folder by pressing Cancel .

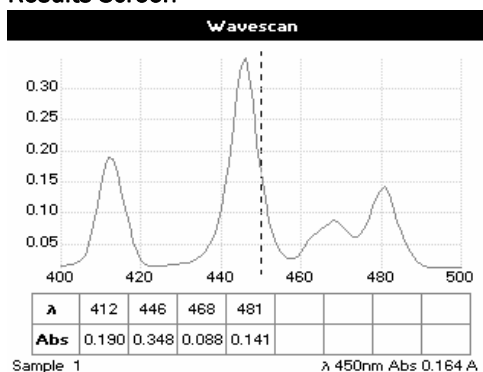
Measurement Screen



Measurement Screen

- Step 8** Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.
- Step 9** Insert sample and press .
- Step 10** Repeat for all samples.

Results Screen



Results Screen

A graph of the wavescan is displayed, along with a table of Absorbance/%T at each peak. Up to eight peaks can be shown. Use the left and right arrows to move the cursor along the graph. When it reaches a peak the peak height and width of the peak is displayed at the top of the screen. To zoom in on the wavelength scale, use the up arrow. This auto-scales on the Absorbance/%T scale (dependent on the Graph Scale option) and this is retained for subsequent measurements. To zoom out again, use the down arrow.

- Step 11** Press **Options** to display available Options which are described next.
- Step 12** Press  and confirm with  to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

1	Parameters...
2	Print
3	Abs/%T
4	Peak Detection...
5	Add Peak...
6	Graph Scale...
7	Sample Number...
8	Save Method...
9	Auto-Print

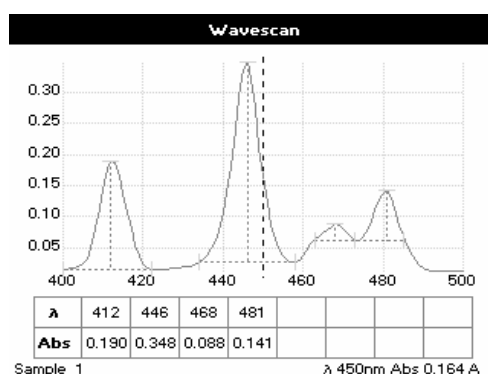
Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle between Absorbance and %T mode.
- 4) Displays Peak Detection Parameter Screen. See description below.
- 5) Manually adds a peak position to the peak table in the results screen at the position set by the cursor. If the cursor is returned to this position the legend "User Defined Peak" is displayed at the top of the scan and this option changes to Delete Peak...
- 6) Displays Graph Scale Parameter Screen. See description below.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (Favourites/Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

Peak Detection (Shortcut button 4)

Peak Detection	
Auto-detect Peaks  Yes 	Peak Detect on Zoom Yes
Min Pk Height 0.01	Sort Peaks By... Wavelength
Min Pk Width 5 nm	Draw Peaks No
 OK	 Cancel



Peak Detection (Shortcut button 4)

Auto Detect Peaks: Turns on and off the automatic peak detection. The following options determine how peaks are detected:

Minimum Peak Height: Minimum height the peak has to be above the higher of the two adjacent minima for the peak to be detected.

Minimum Peak Width: Minimum width of the peak as determined by the difference in wavelength between the highest of the two adjacent minima and the opposing intersection of that higher minimum level and the peak profile. (See the screen displayed below).

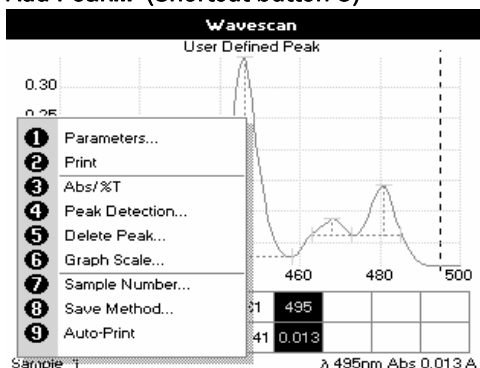
Peak Detect on Zoom: Determines whether peaks are re-assessed and tabulated when the user zooms into a region of the wavescan. If **Off**, leaves the peak detection as determined on the un-zoomed display.

Sort Peaks by...: Determines the sequence that peaks are reported by. Can be wavelength, peak height or peak width.

Draw Peaks: Switches display of peak cursors on and off. These show vertical dashed lines displaying the measured peak height and horizontal dashed lines showing the peak width.

Pressing Cancel  ignores the selection, pressing  accepts them.

Add Peak... (Shortcut button 5)



Add Peak... (Shortcut button 5)

Adds a user defined peak at the current cursor position. The entry is then displayed in inverse colouring to discriminate between user defined peaks and auto-detect peaks. When the cursor is positioned over the user defined peak a legend "User Defined Peak" appears at the top of the graph. The option then changes to Delete Peak to enable the user to remove the peak.

Graph Scale...(Shortcut button 6)

The 'Graph Scale' dialog box is shown. It has a 'Zoom Mode' section with a dropdown menu set to 'y axis'. Below this are two sections: 'x axis limits' and 'y axis limits', both currently set to 'Off'. The 'x axis limits' section has input fields for 'x1' (400 nm) and 'x2' (500 nm). The 'y axis limits' section has input fields for 'y1' (0.00) and 'y2' (2.50). At the bottom are 'OK' and 'Cancel' buttons.

Graph Scale...(Shortcut button 6)

This enables the user to set up a defined graph by defining the limits in either or both of the x and y axes.

Zoom mode:

This sets up the operation of the Zoom keys (up and down arrows). "x & y axis" expands the display around the cursor measurement point, whilst the other options select the absorbance or wavelength axes respectively. With x or y axis limits set to on, zooming out will only be permitted to the set limits.

x/y axis limits:

Setting "x (or y) axis limits" to "On" activates the start and finish points of the desired graph to user defined specific wavelengths and/or absorbance values.

Pressing Cancel  ignores the selection; pressing  accepts it and displays the required graph.

5.4 Kinetics

Simple kinetics studies, where the change in absorbance needs to be followed as a function of time at a fixed wavelength, can be readily performed.

Reagent test kits are routinely used for the enzymatic determination of compounds in food, beverage and clinical laboratories by measuring NAD / NADH conversion at 340 nm. The change in absorbance over a specified time period can be used to provide useful information when an appropriate factor, defined in the reagent kit protocol, is applied. Reaction rate and enzyme activity can be calculated if the factor used takes account of the absorbance difference per unit time, as opposed to the absorbance difference *per se*. For this reason, the change in absorbance per minute ($\Delta A/\text{min}$), concentration ($\Delta A/\text{min} \times \text{factor}$) and **correlation coefficient** (calculated from a best fit of the data points) are displayed. They may not be relevant for simple kinetics experiments.

The procedure to define a new method is as follows:

Parameter Screen

Kinetics - Parameters 1

Wavelength	Delay Time
340 nm	0 Seconds
	Duration
	1 Minute
	Interval
	10 Seconds

Next Cancel

Parameter Screen

- Step 1** Press 3 to select **Functions**.
Step 2 Press 4 to select **Kinetics**
Step 3 **Wavelength**: Enter all numerical values using the keypad numbers or the left and right arrows.
Step 4 **Delay time**: Enter the delay time in seconds before the first measurement is taken. This can be a maximum of 600 seconds (10 minutes).
Step 5 **Duration**: Enter the time in minutes over which measurements are taken. This can be a maximum of 60 minutes.
Step 6 **Interval**: Enter the interval time in seconds between measurements using the left and right arrows. Options are: 5, 10, 20, 30 or 60 seconds.
Step 7 Press Next to go to the next parameters screen OR Press Cancel to return to the **Functions** folder.

Kinetics - Parameters 2

Mode	Pathlength
Delta A	10 mm
Units	
Factor	
1.000	

OK Back

Kinetics - Parameters 2

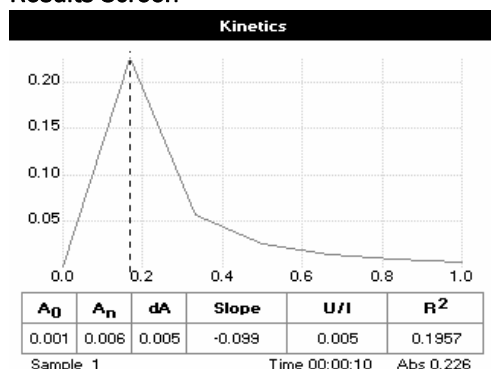
Units

Units
U/l
DP
Auto

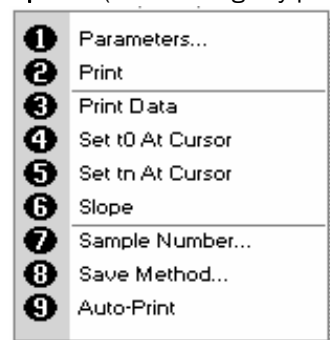
OK Cancel

- Step 8** Select the measurement **Mode** using the left and right arrows.
Delta A: change in absorbance over the measurement duration (or selected period).
Final A: absorbance at the end of the measurement duration (or selected time).
Slope: rate of change of absorbance over the measurement duration or selected period.
Step 9 **Units**: The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows ($\mu\text{g}/\text{ml}$, $\mu\text{g}/\mu\text{l}$, $\text{pmol}/\mu\text{l}$, mg/dl , mmol/l , $\mu\text{mol}/\text{l}$, g/l , mg/l , $\mu\text{g}/\text{l}$, U/l , %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2 Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press OK to store the chosen parameters OR Cancel .
- Step 10** Set the **Factor** by which the result is multiplied to give the amount in the chosen range using the left and right arrows. Range of 0.01 to 9999.
- Step 11** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.
- Step 12** Press Next to enter the Results screen OR Press Cancel to return to the Parameters screen.

Results Screen



Options (select using key pad numbers)



Results Screen

- Step 13** Insert the reference sample and press **Blank** key.
- Step 14** Insert the sample and press **↵** to start the run. Time (min) is displayed at the bottom of the screen, and absorbance data are plotted on the graph as testing proceeds. The table below the graph gives: absorbance values at A₀ (start of calculation), A_n (finish of calculation), dA (change in absorbance), slope, regression parameter (R²) of the calculated slope and the result calculated from the selected parameter.
- Step 15** Use the left and right arrows to move the cursor and display the time and absorbance value at measured data points. Use the up and down arrows to zoom in or out.
- Step 16** Press **Options** to display available Options which are described below.
- Step 17** Press **Escape** **⏮** and confirm with **↵** to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

- 1) Return to parameter screen.
- 2) Print data on the results screen via selected method.
- 3) Print all the data.
- 4) Set the t₀ position (starting point for the slope and dA calculation) at the current cursor position. Value is retained for subsequent samples.
- 5) Set the t_n position (finishing point for the slope and dA calculation) at the current cursor position. Value is retained for subsequent samples.
- 6) Toggle the calculated slope line on and off.
Note: if any data points enclosed by t₀ and t_n are beyond the range of the instrument (> 2.5 A or < -0.3 A) then this option is greyed out.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing **⏮**, or wait.

5.5 Standard Curve

The construction of a multi-point calibration curve from standards of known concentration to quantify unknown samples is a fundamental use of a spectrophotometer; this instrument has the advantage of being able to store this curve as a method, using up to 9 standards.

To include a zero concentration standard, include this in the number of standards to be entered and enter 0.00 for concentration; use a reagent blank when required to enter the zero standard.

The procedure is as follows:

Parameter Screen

Standard Curve - Parameters

Wavelength: 430 nm

Pathlength: 10 mm

Standards: 6

Units: µg/ml

Next Cancel

Parameter Screen

- Step 1** Press 3 to select **Functions**.
- Step 2** Press 5 to select **Standard Curve**
- Step 3** Select the **Wavelength** using the keypad numbers or left and right arrows.
- Step 4** Enter the number of **Standard** concentration points to be used in the curve (1-9).
- Step 5** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.

Standard Curve - Parameters

Units

Units: µg/ml

DP: Auto

OK Cancel

- Step 6 Units:** The user can enter a text string up to 8 characters long. To access a list of pre-defined units press the **Options** key and then use the left/right arrows (µg/ml, µg/µl, pmol/µl, mg/dl, mmol/l, µmol/l, g/l, mg/l, µg/l, U/l, %, ppm, ppb, conc or none). These units can also be edited once OK is pressed. This screen also allows the number of displayed **Decimal Points (DP)** to be selected, from 0 to 2. Note that the result will always be fixed to 5 significant figures regardless of how many decimal points are selected (so 98768.2 will display as 98768 even with 1 decimal point selected). Press OK to store the chosen parameters or Cancel.

Standard Curve - Parameters

Curve Fit: Regression

Calibration: Standards

Replicates: Off

Next Cancel

- Step 7** Select the type of **Curve Fit** using the left and right arrows. Options: straight line regression, a zero regression (this forces the straight line through the origin), interpolated or cubic spline.
- Step 8** Select the **Calibration** mode: either Standards (measure prepared standards) or Manual (keypad data entry) or new standards (using a saved method previous values are blanked, new standard can be measured).
- Step 9** (if standards selected) Select the number of standards to be measured and averaged at each standard concentration point. Can be OFF (1), 2 or 3.
- Step 10** Press Next to enter the Standards screen OR Press Cancel to cancel selections and return to the **Functions** folder.

Standard Screen

Standard Curve - Standards

Std. 1	Std. 4
10.0	40.0
Std. 2	Std. 5
20.0	50.0
Std. 3	Std. 6
30.0	60.0

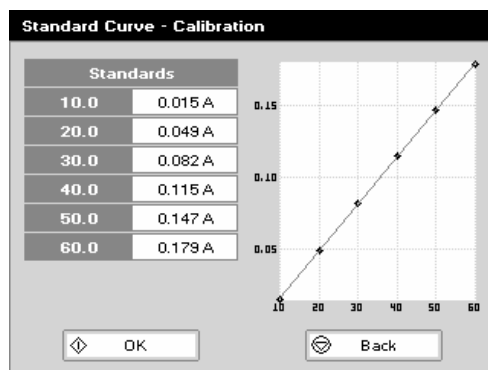
Next Back

Calibration Screen (replicates off)

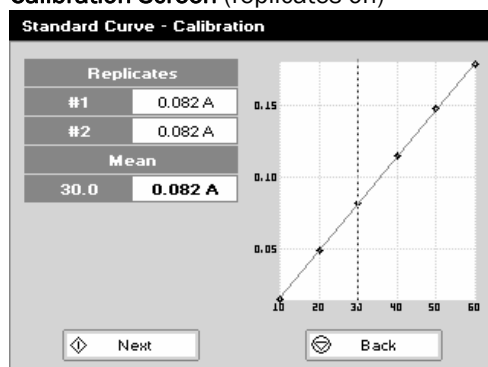
Standard Curve - Calibration

Standards	
10.0	
20.0	
30.0	
40.0	
50.0	
60.0	

Back



Calibration Screen (replicates on)



Standards screen

Step 11 Enter the concentration values by using the keypad numbers and the up and down arrows to move between the different standard boxes. Range 0.001 to 9999.

Step 12 Press Next to enter the Calibration screen. If any duplicate or non-monotonic (increasing entries) are present the unit will beep and highlight the incorrect entry. **OR** Press Back to return to the Parameter screen.

Calibration Screen (replicates off)

Step 13 This shows the calibration values and allows standards to be measured.

Step 14 Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

Step 15 Insert the standard (use C to clear previously stored results before measuring) and press to measure the standard and store the result.

Step 16 Repeat for all standards. A graph will display the results and the fitted curve as the measurements are input.

Step 17 Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Press OK to accept the calibration and go to the Results screen (see below) **OR** Press Back to return to the Standards screen.

Calibration Screen (replicates on)

This shows the calibration values and allows standards to be measured.

Step 18 Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.

Step 19 Press to display the replicate entry boxes. Use C to clear previously stored results before measuring.

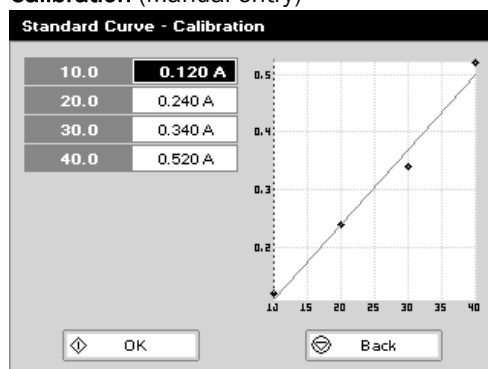
Step 20 Insert the standard and press to measure the standard and store the result.

Step 21 Repeat for all replicates and standards. A graph will display the results and the fitted curve as the measurements are input.

Step 22 Use the up and down arrows to select a standard to be repeated if a poor reading has been obtained. Use C to clear the previous reading.

Step 23 Press to accept the calibration and go to the Results screen (see below) **OR** Press Back to return to the Standards screen.

Calibration (Manual entry)



Calibration (Manual entry)

- Step 24** Shows previously entered calibration values and allows values to be entered via the keypad.
- Step 25** The highlighted box can be edited in order to enter an absorbance value corresponding to a given concentration value using the keypad numbers. Range 0.001 to 9999. Use C to backspace and clear the last digit entered and the up and down arrows to move between boxes.
- Step 26** Press OK to accept the calibration and go to the Results screen (see below) OR Press Back to return to the Standards screen.

Results screen

Standard Curve	
Wavelength 430 nm	Sample 2
Absorbance 0.115 A	Units µg/ml
Curve Fit Regression	Concentration 0.799
Standard Pathlength 10 mm	Sample Pathlength 10 mm

Results screen

- Step 27** Insert the reference sample and press **Blank** key. This will be used for all subsequent samples until changed.
- Step 28** Insert the sample and press . The concentration of the sample is taken and displayed.
- Step 29** Repeat for all samples.
- Step 30** Press **Options** to display available Options which are described below.
- Step 31** Press and confirm with to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

- 1 Parameters...
- 2 Print
- 3 Graph
- 4 Edit Sample Pathlength
- 7 Sample Number...
- 8 Save Method...
- 9 Auto-Print

Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Toggle graph on/off. Displays calibration graph, cursors give values for last measured sample.
- 4) Possibility to edit the sample pathlength.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing , or wait.

5.6 Multiple Wavelength

This makes up to 5 absorbance measurements on the same sample.
The procedure is as follows:

Parameter Screen

Multi Wavelength - Parameters

Wavelengths: 4

Pathlength: 10 mm

OK Cancel

Multi Wavelength - Parameters

λ 1: 300 nm

λ 2: 400 nm

λ 3: 500 nm

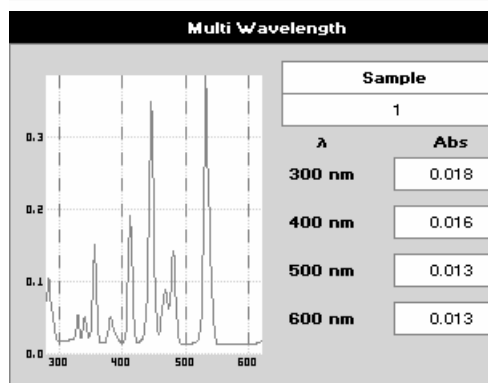
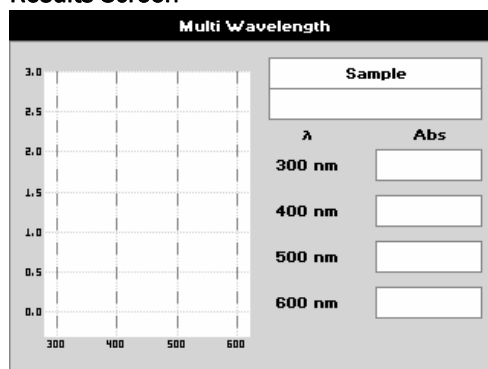
λ 4: 600 nm

OK Cancel

Parameter Screen

- Step 1** Press 3 to select **Functions**.
- Step 2** Press 6 to select **Multi Wavelength**
- Step 3** Select the number of **Wavelengths**.
- Step 4** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.
- Step 5** Press OK to enter the next screen
- Step 6** Enter the **first Wavelength** using either the number keys or the left and right arrows.
- Step 7** Enter the **second Wavelength** as above and repeat for the number of wavelengths selected (up to 5).
- Step 8** Press OK to enter the results screen **OR** Press Cancel to return to the **Functions** folder.

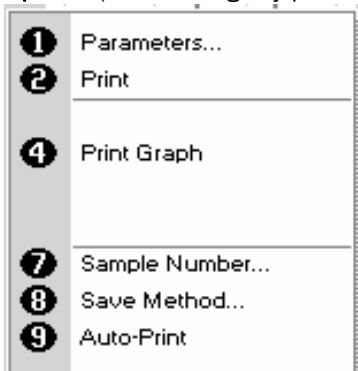
Results Screen



Results Screen

- Step 9** Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.
- Step 10** Insert sample and press .
- Step 11** Repeat for all samples. A scan plot covering the range of wavelengths selected (with cursors at the relevant wavelengths) and a table of values is displayed.
- Step 12** Press **Options** to display available Options which are described below.
- Step 13** Press and confirm with to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)



Options (select using key pad numbers)

- 1) Return to parameters screen
- 2) Print result via selected method.
- 4) Print graph using selected method. It is greyed out if no data are available.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

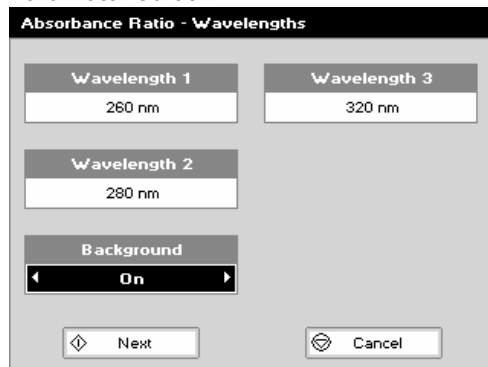
Exit options by pressing , or wait.

5.7 Absorbance Ratio

This makes simple absorbance ratio measurements on samples, measuring the amount of light that has passed through a sample relative to a blank (this can be air) at two wavelengths.

The procedure is as follows:

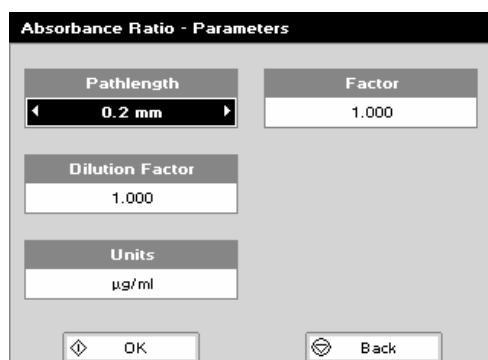
Parameter Screen



The screen is titled "Absorbance Ratio - Wavelengths". It contains three input fields: "Wavelength 1" with the value "260 nm", "Wavelength 2" with the value "280 nm", and "Wavelength 3" with the value "320 nm". Below these is a "Background" section with a toggle switch set to "On". At the bottom are two buttons: "Next" with a right arrow icon and "Cancel" with a left arrow icon.

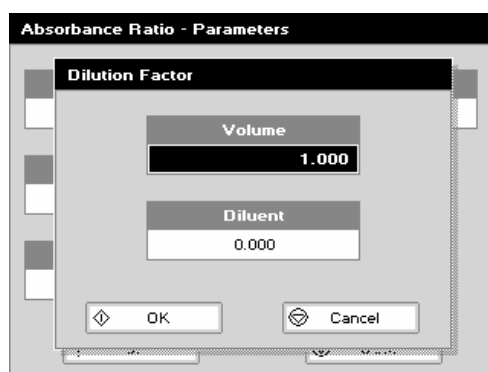
Parameter Screen

- Step 1** Press 3 to select **Functions**.
Step 2 Press 7 to select **Absorbance Ratio**
Step 3 Enter the **first Wavelength** by using the keypad numbers or the left and right arrows.
Step 4 Enter the **second Wavelength** as above.
Step 5 Select whether a **Background** correction is applied to both wavelengths 1 and 2 using the left and right arrows.
Step 6 (If background correction is On) Enter the **third Wavelength**, from which the background correction will be obtained.
Step 7 Press Next  to enter the next screen **OR** Press Cancel  to return to the **Functions** folder.



The screen is titled "Absorbance Ratio - Parameters". It contains three input fields: "Pathlength" with the value "0.2 mm", "Factor" with the value "1.000", and "Dilution Factor" with the value "1.000". Below these is a "Units" section with the value "µg/ml". At the bottom are two buttons: "OK" with a right arrow icon and "Back" with a left arrow icon.

- Step 8** Select the **Pathlength** using the left and right arrows. Options are 0.1, 0.2, 1 and 2 mm for LabelGuard applications and 5 or 10 mm for cuvette applications.
Step 9 (Dilution Factor known) Enter a **Dilution factor** by using the keypad numbers within the range 1.00 – 9999. **OR**



The screen is titled "Absorbance Ratio - Parameters". It shows a sub-screen titled "Dilution Factor" with two input fields: "Volume" with the value "1.000" and "Diluent" with the value "0.000". Below these are two buttons: "OK" with a right arrow icon and "Cancel" with a left arrow icon.

- Step 10** (Calculate Dilution Factor) Press the **Options** key. Enter the **Volume** of the sample (range 0.01 – 9999), using the keypad numbers. Enter the volume of **Diluent** (range 0.01-9999) by using the keypad numbers.
Step 11 Press OK  to calculate the dilution factor and return to the Parameters screen **OR** press Cancel  to cancel selections.
Step 12 Select units of measurement, using left and right arrows. Options are: µg/ml, ng/µl, µg/µl.
Step 13 Enter the factor using the keypad numbers (Range 0.001 to 9999).
Step 14 Press OK  to enter the results screen **OR** Cancel  to return to the **Functions** folder.

Results Screen

Absorbance Ratio	
260 nm	Sample
0.258 A	1
280 nm	Ratio
0.166 A	1.554
Concentration	
12.9 µg/ml	

Results Screen

- Step 15** Insert the reference sample. Press **Blank** key. This will be used for all subsequent samples until changed.
- Step 16** Insert sample and press **↩**.
- Step 17** Repeat for all samples. The absorbance at the selected wavelengths is measured and the ratio between wavelengths 1 and 2 is calculated (both corrected by the background wavelength value if this was selected).
- Step 18** Press **Options** to display available Options which are described below
- Step 19** Press **⏹** and confirm with **↩** to return to the **Functions** folder. Query needs confirmation to avoid unintended escaping the application.

Options (select using key pad numbers)

1	Parameters...
2	Print
3	Graph
7	Sample Number...
8	Save Method...
9	Auto-Print

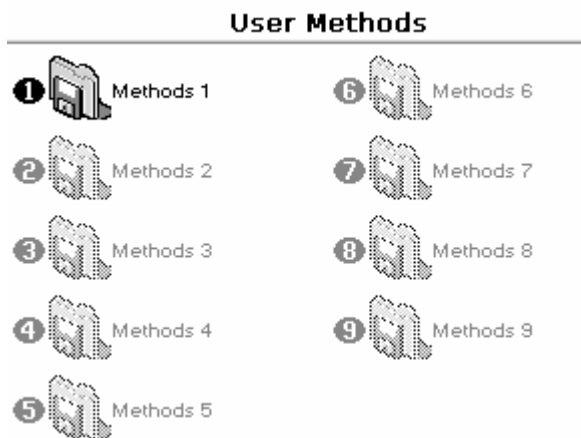
Options (select using key pad numbers)

- 1) Return to parameters screen.
- 2) Print result via selected method.
- 3) Print graph using selected method. It is greyed out if no data are available.
- 7) Sample number – add a prefix to the sample number and reset the incrementing number to the desired value.
- 8) Save method – use the left and right arrows to select a folder to store in (User Methods 1-9), press the down arrow and enter name.
- 9) Auto-print – toggles auto-print on/off.

Exit options by pressing **⏹**, or wait.

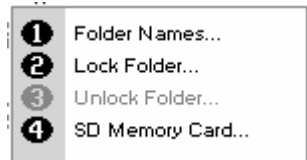
6. USER METHODS

These folders are the storage locations for any user modified Applications (Methods) that are saved in the Options menu. They are accessible from the home folders page. The folder enables the user to quickly select any frequently used Methods. Up to 9 Methods may be stored in the folder.



Folder names can be renamed, locked, unlocked and saved to the SD memory card using the **Options menu**.

Options (select using key pad numbers)



Rename Folder Names

1. Press 1 to select Folder Names.
2. Select the method to be renamed using the left and right arrows.
3. Enter the new name.
4. Press  to save the new name OR  to return to the User Methods folder.

Lock Method

1. Press 2 to select Lock Folder.
2. Select the method to be locked using the left and right arrows.
3. Select a pass code using the keypad numbers or left and right arrows.
4. Press  to lock the method OR  to return to the **User Methods** folder.

Unlock Method

1. Press 3 to select Unlock Folder.
2. Select the method to be unlocked using the left and right arrows.
3. Enter the pass code using the keypad numbers or left and right arrows.
4. Press  to unlock the method OR  to return to the **User Methods** folder

SD Memory Card

Individual or all methods can be copied on the SD Memory Card and can be restored back into the same instrument at a later date. For further details, please refer to the NanoPhotometer Accessory manual.

1. Press 4 to select SD Memory Card.
2. Four options are available:

Backup folder	generates a copy of an individual folder on the SD Memory Card
Restore folder	restores an individual folder from the SD Memory Card to the instrument
Backup all folders	generates a copy of all folders on the SD Memory Card
Restore all folders	restores all folders from the SD Memory Card to the instrument
3. Select the method to be saved using the left and right arrows.
4. Press  to save the method OR  to return to the **User Methods** folder.

Methods - Methods 1

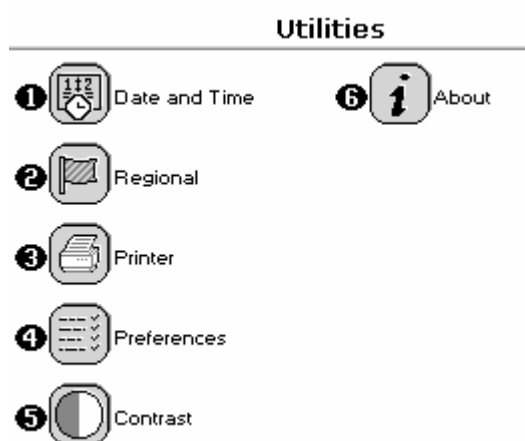


Delete Method

1. Select the method to be deleted using the key pad numbers.
2. Select options and press 1 Delete Method.
3. Press  to delete the method OR  to return to *User Methods* folder.

7. UTILITIES

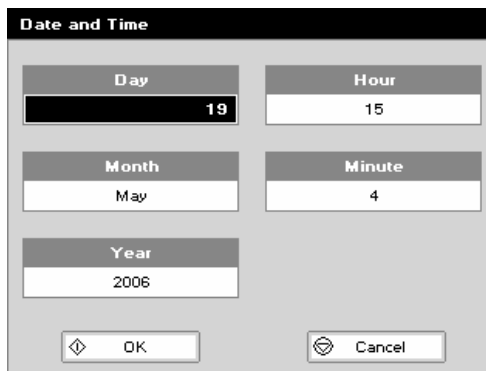
Survey of the available Utilities:



Key pad number	Description
1	Set correct time and date
2	Select preferred number format
3	Printer/output options
4	Select screen layout (themes), history, and Baseline Compensation
5	Adjust screen contrast & brightness
6	Serial number and software version

7.1 Date and Time

The procedure is as follows:



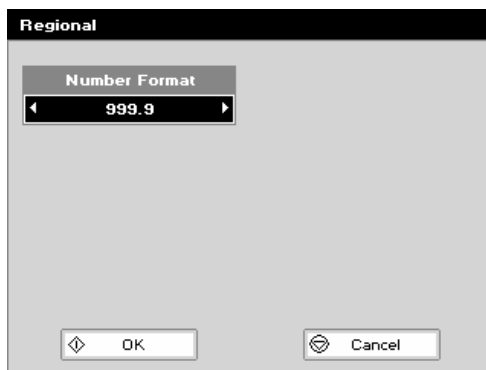
The 'Date and Time' screen displays four input fields: 'Day' (19), 'Hour' (15), 'Month' (May), and 'Minute' (4). Below these is a 'Year' field (2006). At the bottom are 'OK' and 'Cancel' buttons with directional arrow icons.

- Step 1** Enter the **Day** using the keypad numbers or left and right arrows.
- Step 2** Enter the **Month** as above.
- Step 3** Enter the **Year**.
- Step 4** Enter the **Hour**.
- Step 5** Enter the **Minute**. Seconds are zeroed when OK is pressed.
- Step 6** Press OK  to store the settings and return to the **Utilities** folder **OR** Press Cancel  to return to the **Utilities** folder without storing the time.

7.2 Regional

Sets Number Format.

The procedure is as follows:



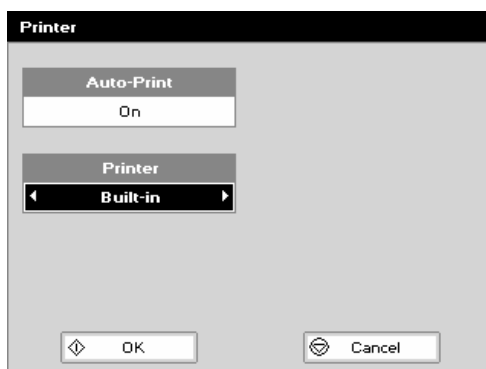
The 'Regional' screen shows a 'Number Format' dropdown menu currently set to '999.9'. At the bottom are 'OK' and 'Cancel' buttons with directional arrow icons.

- Step 1** Set the **Number Format** (decimal point style). Options are “,” or “.”.
- Step 2** Press OK  to store the settings and return to the **Utilities** folder **OR** Press Cancel  to return to the **Utilities** folder without storing the settings.

7.3 Printer

Sets up printing options.

The procedure is as follows:



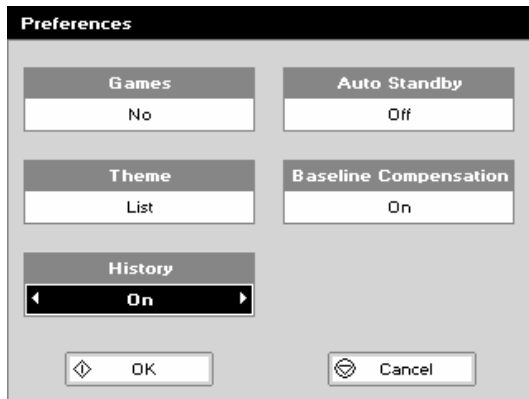
The 'Printer' screen has two settings: 'Auto-Print' (set to 'On') and 'Printer' (set to 'Built-in'). At the bottom are 'OK' and 'Cancel' buttons with directional arrow icons.

- Step 1** Select whether **Auto-print** is on or off using the left and right arrows. When auto-print is on the results are automatically printed after a measurement is taken. When it is off printing has to be initiated manually. This can also be set using the **Options** key in each application or method. The default is OFF.
- Step 2** Select how the data are sent. Options are **Computer USB**  and depending on the attached printer module: **Built-in** , **SD Memory Card**  or **Bluetooth** .
- Step 3** Press OK  to store the settings and return to the **Utilities** folder **OR** Press Cancel  to return to the **Utilities** folder without storing the settings.

7.4 Preferences

Sets user preferences.

The procedure is as follows:

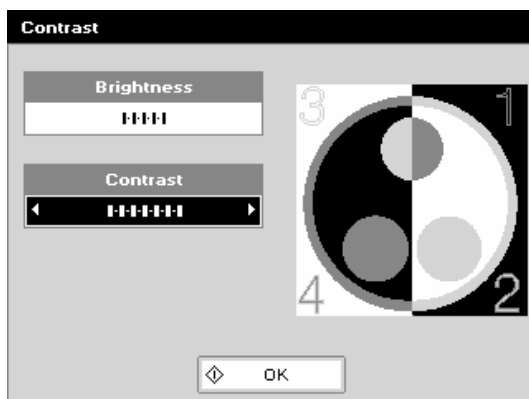


- Step 1** Select **Games** function. This determines whether the games folder is displayed or not; Options are yes or no.
- Step 2** Define the **Screen layout** (Theme) of folders. Options are either a grid format or a list.
- Step 3** Select **History** whether to use previously entered parameters (memory function) or to return to default settings.
- Step 4** Select whether to use a **Standby** mode after defined periods. Options are 1 hour, 2 hours, at night or off.
- Step 5** Select **Baseline Compensation** to improve value stability and to overcome background effects.
- Step 6** Press OK  to store the settings and return to the **Utilities** folder **OR** Press Cancel  to return to the **Utilities** folder without storing the settings.

7.5 Contrast

Ambient temperature can affect the display. This function can optimize the display for local conditions.

The procedure is as follows:



- Step 1** Adjust the **Brightness** using the left and right arrows.
- Step 2** Adjust the **Contrast** using the left and right arrows.
- Step 3** Press OK  to store the settings and return to the **Utilities** folder.

7.6 About



Displays the instrument serial number and software version.

Press OK  to close the window and return to the *Utilities* folder or wait.

8. ACCESSORIES

Certified Didymiumglasfilter for the control of wavelength and photometric accuracy for the NanoPhotometer™	NP-HDF100
NanoPhotometer™ Software Package	NP-SW100
Thermal Paper Printer module for the NanoPhotometer™	B-80-3003-84
Replacement Paper Rolls for Thermal Paper Printer (pack of 5)	VI-G1D-04712
Bluetooth connection module	B-80-3003-96
SD Memory Card Module	B-80-3005-05
Dust cover for the NanoPhotometer™	B-80-3004-02
Lid 5, 2 mm lid for LabelGuard™ Microliter Cell	LG-100-L2
Lid 100, 0.1 mm lid for LabelGuard™ Microliter Cell	LG-100-L01

9. MAINTENANCE

9.1 Maintenance-free Technology

The NanoPhotometer™ technology is maintenance-free. Regular maintenance or calibration is not necessary.

For facilities that are working according to national as well as international guidelines and standards - like Good Laboratory Practice (GLP), Good Manufacturing Practice (GMP) or ISO9000-9004 - the proper performance of a spectrophotometer has to be tested and proved on regular individually set intervals. Implen provides certified NanoPhotometer™ secondary standards as an optional accessory. These NanoPhotometer™ Didymiumglassfilters are suitable for the control and documentation of the wavelength accuracy and the photometric accuracy of your system. Please contact your local Implen office or an authorized Implen partner for further information.

9.2 Lamp Replacement

The xenon lamp should not need replacement until after several years of use. In the unlikely event that it does need replacing, this should be undertaken by a service engineer from your supplier.

9.3 Cleaning and general care of the instrument

External cleaning

Switch off the instrument and disconnect the power cord.

Use a soft damp cloth.

Clean all external surfaces.

A mild liquid detergent may be used to remove stubborn marks.

Changing cell holder or removal for cleaning

This can be removed by undoing the appropriate screws on the bottom of the instrument.

10. SPECIFICATION AND WARRANTY

Technical Specifications:

<i>Wavelength range</i>	190 – 1,100 nm
<i>Wavelength scan range</i>	200 – 950 nm
<i>Sample volume</i>	0,7 to 10 µl with LabelGuard™ Microliter Cell, with cuvettes up to 3,5 ml (quartz or plastic)
<i>Measure time for full scan range</i>	Less than 5 seconds
<i>Wavelength reproducibility</i>	< ± 0.2 nm
<i>Wavelength accuracy</i>	± 2 nm
<i>Spectral Bandwidth</i>	Better than 5 nm
<i>Stray light</i>	< 0.5% at 220 nm using NaI and 340 nm using NaNO ₂
<i>Photometric range</i>	-0.3 – 2.499 A 0-199% T
<i>Detection Range</i>	dsDNA: 2 ng/µl to 8,000 ng/µl, BSA: 0.1 mg/ml to saturation, approx. 160 mg/ml
<i>Photometric reproducibility</i>	±0.003 A (0 to 0.5 A) ±0.007 A (0.5-1.0 A) @ 260 nm
<i>Photometric accuracy</i>	±0.005 A or ±1% of the reading, whichever is the greater
<i>Zero stability</i>	±0.003 A/hour after 20 min warm up @ 340 nm
<i>Noise</i>	0.002 A rms at 0 A @ 260 nm 0.005 A (pk to pk) at 0 A @ 260 nm
<i>Optical arrangement</i>	Dual channel Czerny Turner with flat grating, 1024 pixel CCD array, concave mirrors
<i>Lamp</i>	Xenon flash lamp
<i>Lifetime</i>	10 ⁹ flashes, up to 10 years
<i>Warranty</i>	1 year
<i>Performance verification</i>	Auto diagnostics when switched on
<i>Cell types</i>	15 mm centre height, outside dimension 12.5 mm x 12.5 mm
<i>Cuvette storage</i>	capacity for eight 10 mm cells
<i>Photometric mode</i>	Abs, %T, concentration, scan, ratio, multi wavelength, kinetics in ΔAbs x factor/min
<i>Method storage</i>	Up to 90 methods in user methods
<i>Built-in methods</i>	Nucleid acid, labelling efficiency (nucleic acid, protein), protein and cell density
<i>Display formats</i>	320 x 240 pixels
<i>Size</i>	140 mm x 275 mm x 380 mm
<i>Weight</i>	< 4.5 kg
<i>Operating voltage</i>	90-250 V, 50/60 Hz, Max 30 VA
<i>Input / Output ports</i>	SD Memory Card, USB and Bluetooth for connection to a PC for direct data download for spreadsheet calculations, printout and data storage

Specifications are measured after the instrument has warmed up at a constant ambient temperature and are typical of a production unit. As part of our policy of continuous development, we reserve the right to alter specifications without notice.

Note: Specifications are subject to change without notice

Warranty

- IMPLEN guarantees that the product supplied has been thoroughly tested to ensure that it meets its published specification. The warranty included in the conditions of supply is valid for 12 months only if the product has been used according to the instructions supplied. IMPLEN or your supplier can accept no liability for loss or damage, however caused, arising from the faulty or incorrect use of this product.



11. APPENDIX

11.1 Nucleic acid quantification

For determination nucleic acid concentration in solution the absorbance at wavelength 260 nm is used. The function describing the concentration to absorbance relation is a modification of the Lambert-Beer equation.

$$C_{\text{nuc}} = A_{260} * \text{Factor}_{\text{nuc}} * \text{Lid Factor}$$

C_{nuc}	nucleic acid concentration (ng/μl)
A_{260}	absorbance (AU) of nucleic acids
$\text{Factor}_{\text{nuc}}$	substance specific factor for nucleic acids (ng * cm/μl) (ds DNA 50, ssDNA 37, RNA 40, Oligo 33)
Lid Factor	virtual dilution factor 5, 10, 50 or 100. Dependent on the used LabelGuard Lid.

maximum absorbance wavelength dye-dependent correction factor at 260 nm dye-dependent extinction coefficient

11.2 Nucleic acid fluorescent dye incorporation

To determine the nucleic acid concentration and the dye concentration after probe labelling a modification of the Lambert-Beer equation is used.

- Calculation of the **Nucleic Acid Concentration** with correction of dye contribution to the 260 nm absorbance reading.

$$C_{\text{nuc}} = [A_{260} - (CF_{\text{dye}} * A_{\text{max,dye}})] * \text{Factor}_{\text{nuc}} * \text{Lid factor} * \text{Dilution factor}$$

C_{nuc}	nucleic acid concentration (ng/μl)
A_{260}	absorbance (AU) of nucleic acids
CF_{dye}	dye-dependent correction factor at 260 nm
$A_{\text{max,dye}}$	absorbance at absorption maximum of the dye (AU)
$\text{Factor}_{\text{nuc}}$	substance specific factor for nucleic acids (ng * cm/μl) (ds DNA 50, ssDNA 37, RNA 40, Oligo 33)
Lid Factor	virtual dilution factor 5, 10, 50 or 100. Dependent on the used LabelGuard Lid.

- Calculation of the **Dye concentration**

$$C_{\text{dye}} = (A_{\text{max,dye}} * \text{Lid factor} * \text{Dilution factor}) / (\epsilon_{\text{dye}} * 10^{\text{exp-6}})$$

C_{dye}	dye concentration (pmol/μl)
$A_{\text{max,dye}}$	absorbance at absorption maximum of the dye (AU)
Lid Factor	virtual dilution factor 5, 10, 50 or 100. Dependent on the used LabelGuard Lid.
ϵ_{dye}	dye-dependent extinction coefficient ($\text{M}^{-1} * \text{cm}^{-1}$)

- Calculation of the **Frequency of Incorporation (FOI)** of dye per 1,000 bases:

Formula for dsDNA: $\text{FOI} = (6,49 * A_{\text{max,dye}}) / (\epsilon_{\text{dye}} * 10^{\text{exp-6}} * A_{260,\text{corrected}})$

Formula for ssDNA: $\text{FOI} = (8,77 * A_{\text{max,dye}}) / (\epsilon_{\text{dye}} * 10^{\text{exp-6}} * A_{260,\text{corrected}})$

Formula for RNA: $\text{FOI} = (8,11 * A_{\text{max,dye}}) / (\epsilon_{\text{dye}} * 10^{\text{exp-6}} * A_{260,\text{corrected}})$

Formula for Oligonucleotides: $\text{FOI} = (9,83 * A_{\text{max,dye}}) / (\epsilon_{\text{dye}} * 10^{\text{exp-6}} * A_{260,\text{corrected}})$

$A_{\text{max,dye}}$	absorbance at absorption maximum of the dye (AU)
ϵ_{dye}	dye-dependent extinction coefficient ($\text{M}^{-1} * \text{cm}^{-1}$)
$A_{260, \text{corrected}}$	$A_{260} - (CF_{\text{dye}} * A_{\text{max,dye}})$

The following dye types and parameters are pre-programmed in the NanoPhotometer.

Dye Type	Absorption maximum Dye (nm)	Dye-dependent extinction coefficient ϵ_{dye}	Dye-dependent correction factor (260 nm) CF_{Dye}
Alexa Fluor 350	345	18400	0,25
Alexa Fluor 488	492	62000	0,30
Alexa Fluor 532	525	82300	0,24
Alexa Fluor 546	555	104000	0,21
Alexa Fluor 555	555	150000	0,04
Alexa Fluor 568	576	93000	0,45
Alexa Fluor 594	588	80400	0,43
Alexa Fluor 647	650	239000	0,00
Alexa Fluor 660	660	107000	0,00
Alexa Fluor 680	680	164000	0,00
Cy3	550	150000	0,08
Cy3.5	581	150000	0,08
Cy5	649	250000	0,05
Cy5.5	675	250000	0,05
Oyster-500	503	78000	0,29
Oyster-550	553	150000	0,05
Oyster-556	560	155000	0,03
Oyster-645	649	220000	0,05
Oyster-650	653	200000	0,04
Oyster-656	660	200000	0,04
Texas Red	593	85000	0,23

*In all formulas the molar dye-dependent extinction coefficient is used.

11.3 Protein fluorescent dye incorporation

To determine the protein concentration and the dye concentration after labelling a modification of the Lambert-Beer equation is used.

- Calculation A280 factor

A280 Factor = molecular weight protein / dye-dependent extinction coefficient

- Calculation of the Protein **Concentration** with correction of dye contribution to the 280 nm absorbance reading

$C_{\text{prot}} = [(A_{280} - (CF_{\text{dye}} * A_{\text{max,dye}})) * A_{280} \text{ Factor} * \text{Lid factor} * \text{Dilution factor}]$

C_{prot} protein concentration (mg/ml)
 A_{280} absorbance (AU) of proteins
 CF_{dye} dye-dependent correction factor at 280 nm
 $A_{\text{max,dye}}$ absorbance at absorption maximum of the dye (AU)
 Lid factor virtual dilution factor 5, 10, 50 or 100. Dependent on the used LabelGuard Lid.

- Calculation of the **Dye concentration** [pmol/μl]:

$C_{\text{dye}} = A_{\text{max,dye}} * \text{Lid factor} * \text{Dilution factor} / (\epsilon_{\text{dye}} * 10^{\text{exp-6}})$

C_{dye} dye concentration (pmol/μl)
 $A_{\text{max,dye}}$ absorbance at absorption maximum of the dye (AU)
 Lid factor virtual dilution factor 5, 10, 50 or 100. Dependent on the used LabelGuard Lid.
 ϵ_{dye} dye-dependent extinction coefficient ($\text{M}^{-1} * \text{cm}^{-1}$)

- Calculation of Degree of Labelling (D/P):

$\text{Degree of Labelling} = A_{\text{max,dye}} * \epsilon_{\text{prot}} / [(A_{280} - (A_{\text{max,dye}} * CF_{\text{dye}})) * \epsilon_{\text{dye}}]$

$A_{\text{max,dye}}$ absorbance at absorption maximum of the dye (AU)
 ϵ_{prot} protein-dependent extinction coefficient ($\text{M}^{-1} * \text{cm}^{-1}$)
 CF_{dye} dye-dependent correction factor at 280 nm
 ϵ_{dye} dye-dependent extinction coefficient ($\text{M}^{-1} * \text{cm}^{-1}$)

The following dye types and parameters are pre-programmed in the NanoPhotometer.

Dye Type	Absorption maximum Dyes (nm)	Dye-dependent extinction coefficient ϵ_{dye}	Dye-dependent correction factor (280 nm) CF_{Dye}
Alexa Fluor 350	346	19000	0.19
Alexa Fluor 405	401	34500	0.70
Alexa Fluor 488	495	71000	0.11
Alexa Fluor 647	650	239000	0.03
Cy3	550	150000	0.08
Cy5	649	250000	0.05
DyLight 649	654	250000	0.04
DyLight 488	493	70000	0.15
FITC	495	68000	0.30
Pacific Blue	416	46000	0.20
r-PE	566	200000	0.18
Texas Red	595	80000	0.18

*In all formulas the molar dye-dependent and the molar protein-dependent extinction coefficient is used.